SOLVING A SECOND-ORDER ENERGY-STABLE CAHN–HILLIARD SCHEME BY PRECONDITIONED DESCENT: GLOBAL CONVERGENCE, COMPUTABLE STOPPING RULES, AND ERROR ESTIMATES

[AUTHOR NAMES WITHHELD FOR REVIEW PREPARATION]

Abstract. Second-order, unconditionally energy-stable time discretizations of the Cahn–Hilliard equation are implicit, and in practice their nonlinear step equations are solved only approximately: a few sweeps of a constant-coefficient "linear iteration" are performed and the last iterate is accepted. The stability and accuracy statements available for these schemes, however, refer to the exact solution of each step. We study this gap for the modified Crank–Nicolson Fourier pseudo-spectral scheme of Cheng, Wang, Wise and Yue in its practically used form with an $O(\tau^2)$ stabilization $A(\phi^{n+1} - 2\phi^n + \phi^{n-1})$. We show that each time step is the minimization of a strongly convex functional and that the linear iteration is exactly a fixed-step preconditioned gradient descent for this functional. This yields convergence of the relaxed iteration from any starting value for every stabilization parameter $A \geq 0$, time step and interface width, provided the relaxation parameter respects a bound determined by the maximum norm on a sublevel set; convergence, without a priori knowledge of the sublevel-set bound, of an Armijo line-search variant and of Newton's method with any positive number of preconditioned conjugate-gradient steps; and an explicit local rate. We then prove a two-sided a posteriori bound relating the residual of any candidate to its distance from the exact step solution, and an exact perturbed energy identity valid for any candidate. These give computable acceptance tests, costing no additional transforms, that guarantee decay of a modified energy for every $\tau > 0$ and, with a residual tolerance proportional to τ^2 and under the stated regularity and mesh–time-step conditions, optimal $O(\tau^2 + h^m)$ convergence of the *computed* solution in $\ell^\infty(H^{-1}) \cap \ell^2(H^1)$; in general the error bound separates the discretization error from the computable algebraic term $(\tau \sum_n \|\mathcal{R}_n(\phi^{n+1})\|^2)^{1/2}$. Numerical experiments confirm the theory. The unrelaxed linear iteration can diverge in parameter regimes of practical interest, and relaxation with a fixed number of inner sweeps reduces the observed temporal order, whereas residual-based termination preserves second order. At equal accuracy on a smooth test problem, the computable acceptance rule needs up to 37% fewer transforms than a fixed number of sweeps and 35–67% fewer than a tight tolerance. On a droplet merger and an Ostwald-ripening test, the practical rule with an Armijo inner method reproduces the merger and disappearance steps of a tightly converged Newton–PCG solve with 54–67% fewer transforms.

1. Introduction

We consider the Cahn–Hilliard equation [2] on $\Omega = (0, L)^2$ with periodic boundary conditions,

$$\partial_t \phi = \Delta \mu, \qquad \mu = \phi^3 - \phi - \varepsilon^2 \Delta \phi, \tag{1.1}$$

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the H^{-1} gradient flow of

(1.2)
$$E(\phi)=\int_\Omega\Big(\tfrac14\phi^4-\tfrac12\phi^2\Big)\,dx+\frac{\varepsilon^2}{2}\int_\Omega|\nabla\phi|^2\,dx,\qquad \frac{d}{dt}E(\phi)=-\|\nabla\mu\|_{L^2}^2,\qquad \frac{d}{dt}\int_\Omega\phi\,dx=0.$$

Energy-stable time discretizations are standard: convex splitting [12], secant and discrete-gradient methods [11, 16], second-order convex-splitting schemes [23, 18, 9, 7], stabilized linear schemes [15, 20, 27, 26], and auxiliary-variable formulations [24, 25].

We study the modified Crank–Nicolson Fourier pseudo-spectral scheme of Cheng, Wang, Wise and Yue [7] in its practically used form, with the stabilization $A(\phi^{n+1}-2\phi^n+\phi^{n-1})$ ($\alpha=0$ in [7]). Each step is a cubic system, solved in [7] by a constant-coefficient *linear iteration*; contraction for $\alpha=0$ is left open there [7, Remark 2.10]. Related ℓ^4 contraction results are [29, 4]; preconditioned steepest descent and multigrid are used in [13, 6, 18]. All stability and convergence statements for the scheme concern the *exact* step solution, whereas codes accept an iterate after finitely many sweeps. This paper analyses the scheme *as computed*.

Main results. Let $\mathcal{R}_n$ be the step residual, $\mathcal{J}_n$ the step functional, $\mathcal{B}$ the constant-coefficient preconditioner and $q=\frac32M^2\theta_N$ with M a maximum-norm bound and $\theta_N\le(A+\sqrt{3}\varepsilon\tau^{-1/2})^{-1}$ (Section 2).

(1) $\mathcal{J}_n$ is 1-strongly convex in $\|\cdot\|_{\mathcal{B}^{-1}}$, $\mathcal{R}_n=\nabla\mathcal{J}_n$, and for every candidate v

$$\widetilde{E}(v,\phi^n)-\widetilde{E}(\phi^n,\phi^{n-1})+\mathcal{D}_n(v)=(\mathcal{R}_n(v),v-\phi^n)\qquad\text{(Proposition 2.2 and Theorem 2.5).}$$

(2) The linear iteration is $v^{k+1}=v^k-\mathcal{B}\mathcal{R}_n(v^k)$. With relaxation $\omega<2/(1+q_0)$ it converges globally for all $A\ge0$, $\tau,\varepsilon>0$; with Armijo backtracking, or Newton–PCG with any positive number of conjugate-gradient steps, without a priori knowledge of q_0; locally with factor $\max\{|1-\omega|,|1-\omega(1+q_*)|\}$ (Theorems 3.2, 3.5, 3.7 and 3.9).

(3) $\|v-\phi^{n+1}\|_{\mathcal{B}^{-1}}\le\|\mathcal{R}_n(v)\|_{\mathcal{B}}\le(1+q)\|v-\phi^{n+1}\|_{\mathcal{B}^{-1}}$, with a fully computable q (Corollary 3.11), and the transform-free tests $|(\mathcal{R}_n(v),v-\phi^n)|\le\eta\mathcal{D}_n(v)$, $\|\mathcal{R}_n(v)\|\le c_R\tau^2$ give energy decay of the computed sequence for every $\tau>0$ (Theorems 3.10 and 3.12).

(4) With accepted residuals ρ^n,

$$\|e^n\|_{-1}+\Big(\tau\sum_{j\le n}\varepsilon^2\|\nabla_N e^j\|^2\Big)^{1/2}\le C\Big(\tau^2+h^m+\Big(\tau\sum_{j<n}\|\rho^j\|^2\Big)^{1/2}\Big);$$

hence, under Assumption 4.1, $\tau\le\tau_0$ and $(1+\ln N)^{1/2}(\tau^{3/2}+\tau^{-1/2}h^m)\le c_0'$, $O(\tau^2+h^m)$ under the accuracy test (energy decay needs none of these conditions), and an $O(\tau)$ bound when $\|\rho^n\|=O(\tau)$, which a formal expansion identifies as the regime of a fixed number of relaxed sweeps; for the linearisation at $\phi=0$ the growth rate of K relaxed sweeps is proved to carry the first-order error $-\frac{r}{1-r}\lambda^2\tau$, $r=(1-\omega)^K$, nonzero for $\lambda\ne0$, $0<r<1$ (Theorem 4.3, Corollaries 4.4 and 4.5, and Propositions 3.14 and 3.15).

(5) Experiments (Section 5) confirm divergence of the unrelaxed iteration, loss of order under fixed relaxed sweeps, sharpness of the a posteriori bound, and agreement of the practical rule (Armijo inner method) with tightly converged Newton–PCG in the reported observables at 54–67% fewer transforms on six runs (labyrinthine coarsening, a merger, Ostwald ripening, three droplet patterns).

Relation to preconditioned-descent theory. Preconditioned steepest descent for convex-splitting schemes is analysed in [13, 6]. For the logarithmic (Flory–Huggins) potential and a finite-difference convex-splitting scheme, Diegel, Wang and Wise [10] prove geometric convergence together with positivity of the logarithmic arguments, assuming strict separation of the

discrete solution from ± 1. Park, Salgado and Wise [22] prove linear convergence of preconditioned gradient descent for strongly convex composite functionals when part of the gradient is computed inexactly, with an error term governed by the gradient accuracy. These results concern convergence *within* one implicit step, for other schemes or inexactness models. Here the gradient is exact and the step functional is that of the second-order secant scheme of [7], whose existing linear iteration is the unit-step method. What we add is the treatment of *termination*: a computable two-sided certificate of the algebraic error (Theorem 3.10 and Corollary 3.11), an exact energy identity for arbitrary candidates that turns stopping into a test with guaranteed energy decay of the accepted sequence (Theorems 2.5 and 3.12), and the propagation of the accepted residuals through the time-stepping error analysis, giving the accumulated term $(\tau\sum\|\rho^j\|^2)^{1/2}$ (Theorem 4.3).

Section 2 contains the discrete setting and the variational structure, Section 3 the inner iterations and termination, Section 4 the error analysis, Section 5 the experiments and concluding remarks, and the appendix auxiliary proofs and implementation formulas.

2. Discrete setting and variational structure of a time step

2.1. **Grid functions.** Let $N \geq 2$ be even, $h = L/N$, $x_{ij} = (ih, jh)$, and $\mathcal{V}$ the real L-periodic grid functions. For $f, g \in \mathcal{V}$,

$$(f,g) = h^2 \sum_{i,j=0}^{N-1} f_{ij} g_{ij}, \quad \|f\| = (f,f)^{1/2}, \quad \|f\|_\infty = \max_{i,j} |f_{ij}|, \quad f_{ij} = \sum_{k \in K_N} \hat{f}_k e^{\mathrm{i}\kappa_k \cdot x_{ij}},$$

with $K_N = \{k \in \mathbb{Z}^2 : -\frac{N}{2} < k_1, k_2 \leq \frac{N}{2}\}$, $\kappa_k = 2\pi k/L$, $s_k = |\kappa_k|^2$, and $\|f\|^2 = L^2 \sum_k |\hat{f}_k|^2$. The collocation Laplacian and the associated forms are

$$\widehat{(\Delta_N f)}_k = -s_k \hat{f}_k, \qquad \langle f, g \rangle_1 = (-\Delta_N f, g), \qquad \|\nabla_N f\|^2 = L^2 \sum_k s_k |\hat{f}_k|^2. \tag{2.1}$$

Let $\mathrm{mean}(f) = |\Omega|^{-1}(f,1) = \hat{f}_0$, $\mathring{\mathcal{V}} = \{f : \mathrm{mean}(f) = 0\}$, $\mathsf{P}_0 f = f - \mathrm{mean}(f)$, and on $\mathring{\mathcal{V}}$

$$\|f\|_{-1}^2 = ((-\Delta_N)^{-1} f, f) = L^2 \sum_{k \neq 0} s_k^{-1} |\hat{f}_k|^2. \tag{2.2}$$

Products such as ϕ^3 are evaluated pointwise.

Lemma 2.1. *For $f \in \mathring{\mathcal{V}}$:* (a) $\|f\|^2 \leq \|f\|_{-1} \|\nabla_N f\|$; (b) $\|f\| \leq \frac{L}{2\pi} \|\nabla_N f\|$; (c) $\|f\|_\infty \leq C_N \|\nabla_N f\|$, $C_N = \frac{\sqrt{2}}{\pi}(1 + \ln N)^{1/2}$.

Proof. (a) $\|f\|^2 = L^2 \sum_{k \neq 0} (s_k^{-1/2} |\hat{f}_k|)(s_k^{1/2} |\hat{f}_k|) \leq (L^2 \sum s_k^{-1} |\hat{f}_k|^2)^{1/2} (L^2 \sum s_k |\hat{f}_k|^2)^{1/2}$. (b) $s_k \geq (2\pi/L)^2$ for $k \neq 0$. (c) Appendix A.1. □

2.2. **The scheme.** Let $\tau > 0$, $A \geq 0$ and

$$\chi(a,b) = \tfrac{1}{4}(a^2 + b^2)(a+b), \qquad \chi(a,b)(a-b) = \tfrac{1}{4}(a^4 - b^4). \tag{2.3}$$

Given ϕ^n, ϕ^{n-1} with $\mathrm{mean}(\phi^n) = \mathrm{mean}(\phi^{n-1})$ and $\phi^{-1} := \phi^0$, the scheme [7] ($\alpha = 0$) is

$$\tau^{-1}(\phi^{n+1} - \phi^n) = \Delta_N \mu^{n+1/2}, \tag{2.4a}$$

$$\mu^{n+1/2} = \chi(\phi^{n+1}, \phi^n) - \big(\tfrac{3}{2}\phi^n - \tfrac{1}{2}\phi^{n-1}\big) - \varepsilon^2 \Delta_N \big(\tfrac{3}{4}\phi^{n+1} + \tfrac{1}{4}\phi^{n-1}\big) + A\big(\phi^{n+1} - 2\phi^n + \phi^{n-1}\big). \tag{2.4b}$$

Since $\widehat{(\Delta_N g)}_0 = 0$, $\mathrm{mean}(\phi^{n+1}) = \mathrm{mean}(\phi^n)$. The discrete and modified energies are

(2.5)
$$E_N(\phi) = \big(\tfrac14\phi^4 - \tfrac12\phi^2, 1\big) + \tfrac{\varepsilon^2}{2}\|\nabla_N\phi\|^2, \qquad \widetilde{E}(a,b) = E_N(a) + \big(\tfrac14 + \tfrac{A}{2}\big)\|a-b\|^2 + \tfrac{\varepsilon^2}{8}\|\nabla_N(a-b)\|^2.$$

2.3. **The preconditioner and its metric.** Let $\mathcal{L}_A = \tau^{-1} - A\Delta_N + \frac34\varepsilon^2\Delta_N^2$ and $\mathcal{B} = \mathcal{L}_A^{-1}(-\Delta_N)$ on $\mathring{\mathcal{V}}$, with symbols

$$\lambda_k = \tau^{-1} + As_k + \tfrac34\varepsilon^2 s_k^2, \qquad \beta_k = s_k/\lambda_k \quad (k \neq 0). \tag{2.6}$$

Then $\mathcal{B}$ is symmetric positive definite on $\mathring{\mathcal{V}}$ and, for $f, g \in \mathring{\mathcal{V}}$,

$$\|f\|_{\mathcal{B}^{-1}}^2 = L^2\sum_{k\neq0}\beta_k^{-1}|\hat f_k|^2 = \tfrac{1}{\tau}\|f\|_{-1}^2 + A\|f\|^2 + \tfrac34\varepsilon^2\|\nabla_N f\|^2, \qquad \|g\|_{\mathcal{B}}^2 = L^2\sum_{k\neq0}\beta_k|\hat g_k|^2, \tag{2.7}$$

$$(g,f) \le \|g\|_{\mathcal{B}}\|f\|_{\mathcal{B}^{-1}}, \qquad \|g\|_{\mathcal{B}} = \sup_{0\neq f\in\mathring{\mathcal{V}}}\frac{(g,f)}{\|f\|_{\mathcal{B}^{-1}}}. \tag{2.8}$$

By the arithmetic–geometric mean inequality $\tau^{-1} + \frac34\varepsilon^2 s^2 \ge 2(\frac34\varepsilon^2\tau^{-1})^{1/2}s = \sqrt3\,\varepsilon\tau^{-1/2}s$, with equality at $s^2 = \frac{4}{3\varepsilon^2\tau}$. Hence

$$\theta_N := \max_{k\neq0}\beta_k \le \sup_{s>0}\frac{s}{\tau^{-1} + As + \frac34\varepsilon^2 s^2} = \frac{1}{A + \sqrt3\,\varepsilon\,\tau^{-1/2}}, \tag{2.9}$$

$$\|f\|^2 \le \theta_N\|f\|_{\mathcal{B}^{-1}}^2, \qquad \tfrac34\varepsilon^2\|\nabla_N f\|^2 \le \|f\|_{\mathcal{B}^{-1}}^2, \qquad \tfrac{1}{\tau}\|f\|_{-1}^2 \le \|f\|_{\mathcal{B}^{-1}}^2 \qquad (f\in\mathring{\mathcal{V}}). \tag{2.10}$$

The bound (2.9) is uniform in N.

2.4. **Residual, step functional and convexity.** Fix n, let $\mathcal{V}_n = \{v \in \mathcal{V} : \mathrm{mean}(v) = \mathrm{mean}(\phi^n)\}$, $\tilde\phi^n = \frac32\phi^n - \frac12\phi^{n-1}$, $\ell_n = \tilde\phi^n + A(2\phi^n - \phi^{n-1})$, and for $v \in \mathcal{V}_n$

$$\begin{aligned}\mu_n(v) &= \chi(v,\phi^n) - \tilde\phi^n - \varepsilon^2\Delta_N\big(\tfrac34 v + \tfrac14\phi^{n-1}\big) + A\big(v - 2\phi^n + \phi^{n-1}\big),\\ \mathcal{R}_n(v) &= \tau^{-1}(-\Delta_N)^{-1}(v-\phi^n) + \mathsf{P}_0\mu_n(v) \in \mathring{\mathcal{V}}.\end{aligned} \tag{2.11}$$

Applying $-\Delta_N$ (injective on $\mathring{\mathcal{V}}$),

$$\tau^{-1}(v-\phi^n) = \Delta_N\mu_n(v) - \Delta_N\mathcal{R}_n(v), \tag{2.12}$$

so v solves (2.4) iff $\mathcal{R}_n(v) = 0$. With

(2.13)
$$G(a,b) = \int_0^a \chi(s,b)\,ds = \tfrac{a^4}{16} + \tfrac{a^3b}{12} + \tfrac{a^2b^2}{8} + \tfrac{ab^3}{4}, \quad \chi_a = \partial_a\chi = \tfrac14(3a^2+2ab+b^2) = \tfrac14\big(2a^2 + (a+b)^2\big),$$

define on $\mathcal{V}_n$

$$\mathcal{J}_n(v) = \tfrac{1}{2\tau}\|v-\phi^n\|_{-1}^2 + \big(G(v,\phi^n),1\big) + \tfrac38\varepsilon^2\|\nabla_N v\|^2 + \tfrac{\varepsilon^2}{4}\langle\phi^{n-1},v\rangle_1 + \tfrac{A}{2}\|v\|^2 - (\ell_n, v). \tag{2.14}$$

Proposition 2.2. *Let $v, w \in \mathcal{V}_n$, $d \in \mathring{\mathcal{V}}$ and $j(t) = \mathcal{J}_n(v+td)$.*

(i) $j'(0) = (\mathcal{R}_n(v), d)$ *and* $j''(t) = \|d\|_{\mathcal{B}^{-1}}^2 + (\chi_a(v+td,\phi^n)d, d) \ge \|d\|_{\mathcal{B}^{-1}}^2$.

(ii) $\mathcal{J}_n(w) = \mathcal{J}_n(v) + (\mathcal{R}_n(v), w-v) + \int_0^1(1-t)\,j''_{w-v}(t)\,dt \ge \mathcal{J}_n(v) + (\mathcal{R}_n(v), w-v) + \frac12\|w-v\|_{\mathcal{B}^{-1}}^2$, *where* $j_{w-v}(t) = \mathcal{J}_n(v + t(w-v))$.

(iii) $\mathcal{J}_n$ *has a unique minimizer on* $\mathcal{V}_n$*; it is the unique solution* ϕ^{n+1} *of* (2.4).

(iv) *With* $\mathcal{J}_n^* = \mathcal{J}_n(\phi^{n+1})$,

$$\tfrac12\|v-\phi^{n+1}\|_{\mathcal{B}^{-1}}^2 \le \mathcal{J}_n(v) - \mathcal{J}_n^* \le \tfrac12\|\mathcal{R}_n(v)\|_{\mathcal{B}}^2, \qquad \|v-\phi^{n+1}\|_{\mathcal{B}^{-1}} \le \|\mathcal{R}_n(v)\|_{\mathcal{B}}. \tag{2.15}$$

Proof. (i) Term by term, using $\partial_a G = \chi$, $((-\Delta_N)^{-1}(v-\phi^n), d)$ for the first term and the symmetry of $\langle\cdot,\cdot\rangle_1$,

$$\begin{aligned} j'(0) &= \Big(\tfrac{1}{\tau}(-\Delta_N)^{-1}(v-\phi^n) + \chi(v,\phi^n) - \tfrac{3}{4}\varepsilon^2\Delta_N v - \tfrac{1}{4}\varepsilon^2\Delta_N\phi^{n-1} + Av - \ell_n,\ d\Big) \\ &= \Big(\tfrac{1}{\tau}(-\Delta_N)^{-1}(v-\phi^n) + \mu_n(v), d\Big), \end{aligned}$$

since $-\ell_n + Av = -\tilde{\phi}^n + A(v - 2\phi^n + \phi^{n-1})$; as $\operatorname{mean}(d) = 0$, μ_n may be replaced by $\mathsf{P}_0\mu_n$. Differentiating again, $j''(t) = \frac{1}{\tau}\|d\|_{-1}^2 + \frac{3}{4}\varepsilon^2\|\nabla_N d\|^2 + A\|d\|^2 + (\chi_a(v+td,\phi^n)d, d)$, and $\chi_a \geq 0$ by (2.13). (ii) Taylor's formula $j(1) = j(0) + j'(0) + \int_0^1 (1-t)j''(t)\,dt$ and (i). (iii) By Lemma 2.3, $\mathcal{J}_n(v) \to \infty$ as $\|\nabla_N v\| \to \infty$ in the finite-dimensional affine space $\mathcal{V}_n$; by (ii) $\mathcal{J}_n$ is strictly convex. The minimizer is the unique critical point, $\mathcal{R}_n = 0$, i.e. (2.4). (iv) (ii) with $(v,w) \to (\phi^{n+1}, v)$ gives the left inequality. (ii) with $w = \phi^{n+1}$, (2.8) and $xy \leq \frac{1}{2}x^2 + \frac{1}{2}y^2$ give $\mathcal{J}_n^* \geq \mathcal{J}_n(v) - \|\mathcal{R}_n(v)\|_{\mathcal{B}}\|\phi^{n+1} - v\|_{\mathcal{B}^{-1}} + \frac{1}{2}\|\phi^{n+1} - v\|_{\mathcal{B}^{-1}}^2 \geq \mathcal{J}_n(v) - \frac{1}{2}\|\mathcal{R}_n(v)\|_{\mathcal{B}}^2$. Combining both, $\|v - \phi^{n+1}\|_{\mathcal{B}^{-1}} \leq \|\mathcal{R}_n(v)\|_{\mathcal{B}}$. □

Lemma 2.3. *For $a, b, \ell \in \mathbb{R}$,*

$$G(a,b) - \tfrac{a^4}{32} + \tfrac{b^4}{4} = \Big(\tfrac{a^2}{4\sqrt{2}} + \tfrac{\sqrt{2}\,ab}{6}\Big)^2 + b^2\Big(\tfrac{5}{72}a^2 + \tfrac{1}{4}ab + \tfrac{1}{4}b^2\Big) \geq 0, \qquad \tfrac{a^4}{32} - \ell a \geq -\tfrac{3}{2}|\ell|^{4/3}. \tag{2.16}$$

Consequently, with $C_{J,n} = \big(\frac{1}{4}(\phi^n)^4 + \frac{3}{2}|\ell_n|^{4/3}, 1\big) + \frac{\varepsilon^2}{8}\|\nabla_N\phi^{n-1}\|^2$,

$$\mathcal{J}_n(v) \geq \tfrac{\varepsilon^2}{4}\|\nabla_N v\|^2 - C_{J,n}, \qquad \mathcal{J}_n(v) \leq \bar{J} \ \Rightarrow\ \|v\|_\infty \leq |\operatorname{mean}(\phi^n)| + \tfrac{2C_N}{\varepsilon}\big(\bar{J} + C_{J,n}\big)^{1/2}. \tag{2.17}$$

Proof. Expanding the square, $(\frac{a^2}{4\sqrt{2}} + \frac{\sqrt{2}ab}{6})^2 = \frac{a^4}{32} + \frac{a^3 b}{12} + \frac{a^2 b^2}{18}$, and $\frac{1}{8} - \frac{1}{18} = \frac{5}{72}$, which proves the identity in (2.16); the quadratic form $\frac{5}{72}x^2 + \frac{1}{4}x + \frac{1}{4}$ has discriminant $\frac{1}{16} - \frac{5}{72} = -\frac{1}{144} < 0$. The scalar function $a \mapsto \frac{a^4}{32} - \ell a$ is minimal at $a^3 = 8\ell$, with value $\frac{1}{2}|\ell|^{4/3} - 2|\ell|^{4/3}$. In (2.14), $\frac{\varepsilon^2}{4}\langle\phi^{n-1}, v\rangle_1 \geq -\frac{\varepsilon^2}{8}(\|\nabla_N v\|^2 + \|\nabla_N\phi^{n-1}\|^2)$, the remaining terms are nonnegative, and $\frac{3}{8} - \frac{1}{8} = \frac{1}{4}$. For the last claim apply Lemma 2.1(c) to $v - \operatorname{mean}(v)$. □

2.5. **Secant calculus and a spectral bound.** For $a, b, c \in \mathbb{R}$,

$$\begin{aligned} \chi(a,b) - \chi(c,b) &= D(a,c;b)(a-c), \\ D(a,c;b) &= \tfrac{1}{4}\big(a^2 + ac + c^2 + b(a+c) + b^2\big) = \tfrac{1}{8}\big[(a+c)^2 + (a+b)^2 + (b+c)^2\big], \end{aligned} \tag{2.18}$$

which follows from $a^3 - c^3 = (a-c)(a^2+ac+c^2)$ and $a^2 - c^2 = (a-c)(a+c)$. If $|a|, |b|, |c| \leq M$, then

$$0 \leq D(a,c;b) \leq \tfrac{1}{8}\cdot 3\cdot(2M)^2 = \tfrac{3}{2}M^2, \qquad 0 \leq \chi_a(a,b) = D(a,a;b) \leq \tfrac{3}{2}M^2. \tag{2.19}$$

Set

$$q(M) = \tfrac{3}{2}M^2\theta_N. \tag{2.20}$$

Lemma 2.4. *Let $0 \leq D \in \mathcal{V}$ with $\|D\|_\infty \leq \frac{3}{2}M^2$ and $T = \mathcal{B}\mathsf{P}_0 D : \mathring{\mathcal{V}} \to \mathring{\mathcal{V}}$. Then T is self-adjoint and positive semi-definite in $(\mathcal{B}^{-1}\cdot,\cdot)$, its spectrum lies in $[0, q(M)]$, and for every polynomial p*

$$\|p(T)u\|_{\mathcal{B}^{-1}} \leq \max_{\lambda\in[0,q(M)]} |p(\lambda)|\,\|u\|_{\mathcal{B}^{-1}} \qquad (u \in \mathring{\mathcal{V}}).$$

Proof. For $u, w \in \mathring{\mathcal{V}}$, $(\mathcal{B}^{-1}Tu, w) = (\mathsf{P}_0 Du, w) = (Du, w)$, which is symmetric, and $0 \leq (Du, u) \leq \frac{3}{2}M^2\|u\|^2 \leq q(M)\|u\|_{\mathcal{B}^{-1}}^2$ by (2.10). The spectral theorem in $(\mathring{\mathcal{V}}, (\mathcal{B}^{-1}\cdot,\cdot))$ gives the rest. □

For $w \in \mathcal{V}_n$ and $d \in \mathring{\mathcal{V}}$ with $\|w\|_\infty, \|\phi^n\|_\infty \le M$, Proposition 2.2(i), (2.19) and Lemma 2.4 give

$$\|d\|_{\mathcal{B}^{-1}}^2 \le \frac{d^2}{dt^2}\mathcal{J}_n(w+td)\Big|_{t=0} \le \big(1+q(M)\big)\|d\|_{\mathcal{B}^{-1}}^2. \tag{2.21}$$

2.6. **An energy identity for arbitrary candidates.**

Theorem 2.5. *For $v \in \mathcal{V}_n$ let $\delta = v-\phi^n$, $\delta_0 = \phi^n - \phi^{n-1}$, $\delta_2 = \delta - \delta_0$ and*

$$\mathcal{D}_n(v) = \tfrac{1}{\tau}\|\delta\|_{-1}^2 + \big(\tfrac14 + \tfrac{A}{2}\big)\|\delta_2\|^2 + \tfrac{\varepsilon^2}{8}\|\nabla_N\delta_2\|^2 \ge 0. \tag{2.22}$$

Then

$$\widetilde{E}(v,\phi^n) - \widetilde{E}(\phi^n,\phi^{n-1}) + \mathcal{D}_n(v) = (\mathcal{R}_n(v), v-\phi^n). \tag{2.23}$$

Proof. Write $a=v$, $b=\phi^n$, $c=\phi^{n-1}$, so $\delta = a-b$, $\delta_0 = b-c$, $\delta_2 = a-2b+c$. By (2.11) and $\mathrm{mean}(\delta)=0$, $(\mathcal{R}_n(v),\delta) = \frac{1}{\tau}\|\delta\|_{-1}^2 + (\mu_n(v),\delta)$. With the polarization identities

$$(x-y,x) = \tfrac12\big(\|x\|^2 - \|y\|^2 + \|x-y\|^2\big), \qquad (x+y,x-y) = \|x\|^2 - \|y\|^2,$$

and their analogues for $\langle\cdot,\cdot\rangle_1$, the four parts of $\mu_n(v)$ give

$$\begin{aligned}
(\chi(a,b), a-b) &= \big(\tfrac14 a^4 - \tfrac14 b^4, 1\big),\\
-(\tfrac32 b - \tfrac12 c, a-b) &= -\tfrac12(a+b,a-b) + \tfrac12(\delta_2,\delta)\\
&= -\tfrac12\big(\|a\|^2 - \|b\|^2\big) + \tfrac14\big(\|\delta\|^2 - \|\delta_0\|^2 + \|\delta_2\|^2\big),\\
\varepsilon^2\big\langle \tfrac34 a + \tfrac14 c, a-b\big\rangle_1 &= \varepsilon^2\big\langle \tfrac12(a+b) + \tfrac14\delta_2, a-b\big\rangle_1\\
&= \tfrac{\varepsilon^2}{2}\big(\|\nabla_N a\|^2 - \|\nabla_N b\|^2\big) + \tfrac{\varepsilon^2}{8}\big(\|\nabla_N\delta\|^2 - \|\nabla_N\delta_0\|^2 + \|\nabla_N\delta_2\|^2\big),\\
A(\delta_2,\delta) &= \tfrac{A}{2}\big(\|\delta\|^2 - \|\delta_0\|^2 + \|\delta_2\|^2\big),
\end{aligned}$$

where $\delta_2 = \delta - \delta_0$ was used in the form $(\delta_2,\delta) = \frac12(\|\delta\|^2 - \|\delta_0\|^2 + \|\delta_2\|^2)$, and $\frac34 a + \frac14 c = \frac12(a+b) + \frac14(a-2b+c)$. Summing, $(\mu_n(v),\delta) = E_N(a) - E_N(b) + (\frac14 + \frac{A}{2})(\|\delta\|^2 - \|\delta_0\|^2) + \frac{\varepsilon^2}{8}(\|\nabla_N\delta\|^2 - \|\nabla_N\delta_0\|^2) + (\frac14+\frac{A}{2})\|\delta_2\|^2 + \frac{\varepsilon^2}{8}\|\nabla_N\delta_2\|^2 = \widetilde{E}(a,b) - \widetilde{E}(b,c) + \mathcal{D}_n(v) - \frac{1}{\tau}\|\delta\|_{-1}^2$. □

Corollary 2.6. *$\widetilde{E}(\phi^{n+1},\phi^n) + \mathcal{D}_n(\phi^{n+1}) = \widetilde{E}(\phi^n,\phi^{n-1})$, and $\mathcal{D}_n(\phi^{n+1}) = 0$ iff $\phi^{n+1} = \phi^n = \phi^{n-1}$.*

Proof. $\mathcal{R}_n(\phi^{n+1}) = 0$ in (2.23). If $\mathcal{D}_n = 0$ then $\delta = 0$ (since $\|\cdot\|_{-1}$ is a norm on $\mathring{\mathcal{V}}$) and $\delta_2 = 0$, hence $\delta_0 = \delta - \delta_2 = 0$. □

3. Inner iterations and computable termination

3.1. **The linear iteration is preconditioned gradient descent.** Let $f_n = \tau^{-1}\phi^n - \Delta_N\ell_n - \frac14\varepsilon^2\Delta_N^2\phi^{n-1}$ and, for $v\in\mathcal{V}_n$, $\mathcal{L}_A\mathcal{G}_n(v) = f_n + \Delta_N\chi(v,\phi^n)$, i.e.

$$\widehat{\mathcal{G}_n(v)}_k = \lambda_k^{-1}\big(\hat f_{n,k} - s_k\,\widehat{\chi(v,\phi^n)}_k\big). \tag{3.1}$$

The linear iteration of [7] is $v\mapsto\mathcal{G}_n(v)$; its relaxed form is

$$v^{k+1} = (1-\omega_k)v^k + \omega_k\mathcal{G}_n(v^k), \qquad v^0 = 2\phi^n - \phi^{n-1}. \tag{3.2}$$

Lemma 3.1. *For $v\in\mathcal{V}_n$: $\mathcal{G}_n(v)\in\mathcal{V}_n$, $v-\mathcal{G}_n(v) = \mathcal{B}\mathcal{R}_n(v)$, and (3.2) is*

$$v^{k+1} = v^k - \omega_k\mathcal{B}\mathcal{R}_n(v^k), \tag{3.3}$$

gradient descent for $\mathcal{J}_n$ in $(\mathring{\mathcal{V}}, (\mathcal{B}^{-1}\cdot,\cdot))$: $(\mathcal{B}^{-1}(\mathcal{B}\mathcal{R}_n(v)), d) = \frac{d}{dt}\mathcal{J}_n(v+td)|_{t=0}$.

Proof. Since $\Delta_N \mathsf{P}_0 = \Delta_N$, (2.11) gives

$$\begin{aligned}-\Delta_N \mathcal{R}_n(v) &= \tfrac{1}{\tau}(v-\phi^n) - \Delta_N \chi(v,\phi^n) + \Delta_N \tilde{\phi}^n + \tfrac{3}{4}\varepsilon^2 \Delta_N^2 v + \tfrac{1}{4}\varepsilon^2 \Delta_N^2 \phi^{n-1} \\ &\quad - A\Delta_N (v - 2\phi^n + \phi^{n-1}) \\ &= \mathcal{L}_A v - \big(\tfrac{1}{\tau}\phi^n - \Delta_N \ell_n - \tfrac{1}{4}\varepsilon^2 \Delta_N^2 \phi^{n-1}\big) - \Delta_N \chi(v,\phi^n) = \mathcal{L}_A\big(v - \mathcal{G}_n(v)\big).\end{aligned}$$

The $k=0$ component gives $\tau^{-1}(\hat{v}_0 - \widehat{\mathcal{G}_n(v)}_0) = 0$, so $v - \mathcal{G}_n(v) \in \mathring{\mathcal{V}}$ and $v - \mathcal{G}_n(v) = \mathcal{L}_A^{-1}(-\Delta_N)\mathcal{R}_n(v) = \mathcal{B}\mathcal{R}_n(v)$. The last statement is Proposition 2.2(i). □

With $\hat{p} = \hat{v} - \widehat{\mathcal{G}_n(v)}$, $\widehat{\mathcal{R}_n(v)}_k = \beta_k^{-1}\hat{p}_k$ and, by Parseval,

$$\|\mathcal{R}_n(v)\|_{\mathcal{B}}^2 = \|p\|_{\mathcal{B}^{-1}}^2 = L^2 \sum_{k\neq 0} \beta_k^{-1} |\hat{p}_k|^2, \qquad \|\mathcal{R}_n(v)\|^2 = L^2 \sum_{k\neq 0} \beta_k^{-2} |\hat{p}_k|^2. \tag{3.4}$$

A sweep costs one forward transform (of $\chi(v,\phi^n)$) and one inverse transform; every quantity used below is evaluated from Fourier coefficients (Appendix A.6).

3.2. **Global convergence.**

Theorem 3.2. *Let $v^0 \in \mathcal{V}_n$, $S_0 = \{v \in \mathcal{V}_n : \mathcal{J}_n(v) \le \mathcal{J}_n(v^0)\}$, $q_0 = q(M_0)$ with $M_0 = \max\{\|\phi^n\|_\infty, \max_{S_0}\|v\|_\infty\}$, finite by Lemma 2.3. If $\omega_k = \omega \in (0, \frac{2}{1+q_0})$, then $v^k \in S_0$ and*

$$\mathcal{J}_n(v^{k+1}) \le \mathcal{J}_n(v^k) - \omega\big(1 - \tfrac{\omega(1+q_0)}{2}\big)\|\mathcal{R}_n(v^k)\|_{\mathcal{B}}^2, \tag{3.5}$$

$$\mathcal{J}_n(v^k) - \mathcal{J}_n^* \le \kappa(\omega)^k \big(\mathcal{J}_n(v^0) - \mathcal{J}_n^*\big), \qquad \kappa(\omega) = 1 - \omega\big(2 - \omega(1+q_0)\big), \tag{3.6}$$

$$\|v^k - \phi^{n+1}\|_{\mathcal{B}^{-1}} \le \kappa(\omega)^{k/2} \|\mathcal{R}_n(v^0)\|_{\mathcal{B}}, \qquad \min_\omega \kappa = \kappa\big(\tfrac{1}{1+q_0}\big) = \tfrac{q_0}{1+q_0}. \tag{3.7}$$

Proof. Let $v = v^k \in S_0$, $p = \mathcal{B}\mathcal{R}_n(v) \neq 0$, $g(t) = \mathcal{J}_n(v - tp)$, $g'(0) = -(\mathcal{R}_n(v), \mathcal{B}\mathcal{R}_n(v)) = -\|\mathcal{R}_n(v)\|_{\mathcal{B}}^2 = -\|p\|_{\mathcal{B}^{-1}}^2$. Let $t^* = \sup\{t \le \omega : \ v - sp \in S_0 \ \forall s \in [0,t]\} > 0$. On $[0,t^*]$, (2.21) with $M = M_0$ gives $g'' \le (1+q_0)\|p\|_{\mathcal{B}^{-1}}^2$, so

$$g(t) = g(0) + tg'(0) + \int_0^t (t-s) g''(s)\, ds \le g(0) - t\big(1 - \tfrac{t(1+q_0)}{2}\big)\|p\|_{\mathcal{B}^{-1}}^2 < g(0) \qquad (0 < t \le t^*).$$

If $t^* < \omega$, continuity of g contradicts maximality of t^*. Hence $t^* = \omega$, $v^{k+1} \in S_0$ and (3.5). By (2.15), $\|\mathcal{R}_n(v^k)\|_{\mathcal{B}}^2 \ge 2(\mathcal{J}_n(v^k) - \mathcal{J}_n^*)$, and (3.5) yields $\mathcal{J}_n(v^{k+1}) - \mathcal{J}_n^* \le [1 - 2\omega(1 - \frac{\omega(1+q_0)}{2})](\mathcal{J}_n(v^k) - \mathcal{J}_n^*) = \kappa(\omega)(\mathcal{J}_n(v^k) - \mathcal{J}_n^*)$. Then $\frac{1}{2}\|v^k - \phi^{n+1}\|_{\mathcal{B}^{-1}}^2 \le \mathcal{J}_n(v^k) - \mathcal{J}_n^* \le \kappa^k(\mathcal{J}_n(v^0) - \mathcal{J}_n^*) \le \frac{1}{2}\kappa^k \|\mathcal{R}_n(v^0)\|_{\mathcal{B}}^2$. Finally $\kappa'(\omega) = -2 + 2\omega(1+q_0) = 0$ at $\omega = \frac{1}{1+q_0}$, $\kappa = 1 - \frac{1}{1+q_0}$. □

Corollary 3.3. *For $\omega = 1$, $\mathcal{J}_n(v^k) - \mathcal{J}_n^* \le q_0^k(\mathcal{J}_n(v^0) - \mathcal{J}_n^*)$ whenever $q_0 < 1$, in particular whenever*

$$\tfrac{3}{2}M_0^2 < A + \sqrt{3}\,\varepsilon\,\tau^{-1/2} \qquad (\Rightarrow\ q_0 = \tfrac{3}{2}M_0^2\theta_N < 1 \text{ by (2.9)}). \tag{3.8}$$

For $A = 0$ this is $\tau < \frac{4}{3}\varepsilon^2 M_0^{-4}$.

Proof. $\kappa(1) = 1 - (2 - 1 - q_0) = q_0$. □

Lemma 3.4. *For $v \in \mathcal{V}_n$, $d \in \mathring{\mathcal{V}}$, with $b = \phi^n$,*

$$\mathcal{J}_n(v+d) - \mathcal{J}_n(v) = (\mathcal{R}_n(v), d) + \tfrac{1}{2}\|d\|_{\mathcal{B}^{-1}}^2 + \Big(d^2\big(\tfrac{1}{2}\chi_a(v,b) + d\,(\tfrac{3v+b}{12} + \tfrac{d}{16})\big), 1\Big). \tag{3.9}$$

Proof. The quadratic part of $\mathcal{J}_n$ contributes $(\tau^{-1}(-\Delta_N)^{-1}(v-\phi^n) - \frac{3}{4}\varepsilon^2\Delta_N v - \frac{1}{4}\varepsilon^2\Delta_N\phi^{n-1} + Av - \ell_n, d) + \frac{1}{2}\|d\|^2_{\mathcal{B}^{-1}}$. Since $G(\cdot,b)$ is a quartic polynomial, $G(v+d,b) - G(v,b) = \chi d + \frac{1}{2}\chi_a d^2 + \frac{1}{6}\chi_{aa}d^3 + \frac{1}{24}\chi_{aaa}d^4$ with $\chi_{aa} = \frac{3a+b}{2}$, $\chi_{aaa} = \frac{3}{2}$. Adding and using Proposition 2.2(i) for the linear term gives (3.9). □

Formula (3.9) avoids the cancellation in $\mathcal{J}_n(v+d) - \mathcal{J}_n(v)$, where both values are $O(1)$ and their difference is $O(\|\mathcal{R}_n\|^2_{\mathcal{B}})$.

Theorem 3.5 (Armijo backtracking)**.** *Let $c \in (0, \frac{1}{2})$, $\gamma \in (0,1)$, $\hat\omega_k \in (0,1]$ with $\hat\omega_k \geq \min\{\hat\omega_0, \omega_{k-1}\}$, and let ω_k be the largest $\omega = \hat\omega_k\gamma^j$, $j \geq 0$, with*

$$\mathcal{J}_n\big(v^k - \omega\mathcal{B}\mathcal{R}_n(v^k)\big) \leq \mathcal{J}_n(v^k) - c\,\omega\|\mathcal{R}_n(v^k)\|^2_{\mathcal{B}}.$$

Then

$$\omega_k \geq \underline{\omega} = \min\Big\{\hat\omega_0, \frac{2\gamma(1-c)}{1+q_0}\Big\}, \qquad \mathcal{J}_n(v^{k+1}) - \mathcal{J}_n^* \leq (1 - 2c\,\underline{\omega})\big(\mathcal{J}_n(v^k) - \mathcal{J}_n^*\big).$$

Proof. By the proof of Theorem 3.2, every $\omega \in (0, \frac{2}{1+q_0})$ gives $\mathcal{J}_n(v^k - \omega p) \leq \mathcal{J}_n(v^k) - \omega(1 - \frac{\omega(1+q_0)}{2})\|\mathcal{R}_n(v^k)\|^2_{\mathcal{B}}$, and $1 - \frac{\omega(1+q_0)}{2} \geq c$ iff $\omega \leq \frac{2(1-c)}{1+q_0}$. Backtracking therefore stops at some $\hat\omega_k\gamma^j \geq \min\{\hat\omega_k, \gamma\frac{2(1-c)}{1+q_0}\}$, and $\omega_k \geq \underline{\omega}$ by induction. With (2.15), $\mathcal{J}_n(v^k) - \mathcal{J}_n(v^{k+1}) \geq c\,\omega_k\|\mathcal{R}_n(v^k)\|^2_{\mathcal{B}} \geq 2c\,\underline{\omega}(\mathcal{J}_n(v^k) - \mathcal{J}_n^*)$. □

Remark 3.6. For $p = \mathcal{B}\mathcal{R}_n(v)$ and $b = \phi^n$, $(\mathcal{R}_n(v), p) = \|\mathcal{R}_n(v)\|^2_{\mathcal{B}} = \|p\|^2_{\mathcal{B}^{-1}}$ gives

$$\mathcal{J}_n(v-p) - \mathcal{J}_n(v) = -\tfrac{1}{2}\|\mathcal{R}_n(v)\|^2_{\mathcal{B}} + \big(G(v-p,b) - G(v,b) + \chi(v,b)\,p, 1\big),$$

and the remainder is ≥ 0 because $G(\cdot,b)$ is convex ($\partial_a^2 G = \chi_a \geq 0$). Hence $c = \frac{1}{2}$ accepts the unit step only if the remainder vanishes. We use $c = \frac{1}{4}$, $\gamma = \frac{1}{2}$, $\hat\omega_k = \min\{1, 2\omega_{k-1}\}$.

3.3. **Local rate.**

Theorem 3.7. *Let $\omega \in (0,2)$, $\omega_k = \omega$, $\xi^k = v^k - \phi^{n+1}$, $M_k = \max\{\|v^k\|_\infty, \|\phi^{n+1}\|_\infty, \|\phi^n\|_\infty\}$, $q_k = q(M_k)$. Then*

$$\|\xi^{k+1}\|_{\mathcal{B}^{-1}} \leq r(\omega, q_k)\|\xi^k\|_{\mathcal{B}^{-1}}, \qquad r(\omega,q) = \max\{|1-\omega|, |1-\omega(1+q)|\}, \tag{3.10}$$

and if $v^k \to \phi^{n+1}$, $\limsup_k \|\xi^{k+1}\|_{\mathcal{B}^{-1}}/\|\xi^k\|_{\mathcal{B}^{-1}} \leq r(\omega, q_*)$, $q_* = q(\max\{\|\phi^{n+1}\|_\infty, \|\phi^n\|_\infty\})$, with $\min_\omega r(\omega, q_*) = r(\frac{2}{2+q_*}, q_*) = \frac{q_*}{2+q_*}$ and $r(1, q_*) = q_*$.

Proof. By (2.18) and $\mathcal{R}_n(\phi^{n+1}) = 0$, $\mathcal{R}_n(v^k) = \mathcal{R}_n(v^k) - \mathcal{R}_n(\phi^{n+1}) = \mathcal{B}^{-1}\xi^k + \mathsf{P}_0 D_k\xi^k$ with $D_k = D(v^k, \phi^{n+1}; \phi^n)$, because $\mathcal{R}_n$ is affine except for $\mathsf{P}_0\chi(\cdot,\phi^n)$. Hence

$$\xi^{k+1} = \xi^k - \omega\mathcal{B}\mathcal{R}_n(v^k) = \big[(1-\omega)I - \omega T_k\big]\xi^k, \qquad T_k = \mathcal{B}\mathsf{P}_0 D_k. \tag{3.11}$$

Lemma 2.4 with $p(\lambda) = 1 - \omega - \omega\lambda$ gives (3.10), since $\max_{[0,q]}|1-\omega-\omega\lambda| = r(\omega,q)$. $M_k \to \max\{\|\phi^{n+1}\|_\infty, \|\phi^n\|_\infty\}$ gives the limit; $r(\cdot, q_*)$ is minimal where $1-\omega = -(1-\omega(1+q_*))$, i.e. $\omega = \frac{2}{2+q_*}$. □

3.4. **Newton–PCG with truncated conjugate gradients.**

At $v \in \mathcal{V}_n$ let $g = \mathcal{B}\mathcal{R}_n(v)$ and $H = I + \mathcal{B}\mathsf{P}_0\chi_a(v,\phi^n)$, so that the Newton equation $(\mathcal{B}^{-1} + \mathsf{P}_0\chi_a)d = -\mathcal{R}_n(v)$ reads $Hd = -g$. With $\tilde q = q(\max\{\|v\|_\infty, \|\phi^n\|_\infty\})$, H is self-adjoint in $(\mathcal{B}^{-1}\cdot,\cdot)$ with spectrum in $[1, 1+\tilde q]$ (Lemma 2.4). Conjugate gradients for $Hd = -g$ in this inner product, started from $d = 0$, is PCG with preconditioner $\mathcal{B}$; one iteration costs one transform pair.

Lemma 3.8. *Let d be the j-th CG iterate, $j \ge 1$, and write $\langle\cdot,\cdot\rangle = (\mathcal{B}^{-1}\cdot,\cdot)$, $|\cdot| = \|\cdot\|_{\mathcal{B}^{-1}}$. Then*

$$\langle g,d\rangle \le -|d|^2, \qquad -\langle g,d\rangle \ge \frac{|g|^2}{1+\tilde q}, \qquad |d| \le |g|, \qquad \frac{-\langle g,d\rangle}{|g|\,|d|} \ge \frac{1}{1+\tilde q}. \tag{3.12}$$

Proof. d minimizes $m(w) = \langle g,w\rangle + \frac12\langle Hw,w\rangle$ over the Krylov space $\mathcal{K}_j \ni g$, so $\langle Hd+g,w\rangle = 0$ for $w \in \mathcal{K}_j$. With $w = d$: $\langle g,d\rangle = -\langle Hd,d\rangle \le -|d|^2$, hence also $|d|^2 \le |g||d|$. Further, since $-tg \in \mathcal{K}_j$, $m(d) = \frac12\langle g,d\rangle \le \min_t m(-tg) \le \min_t\big(-t|g|^2 + \frac{t^2}{2}(1+\tilde q)|g|^2\big) = -\frac{|g|^2}{2(1+\tilde q)}$. The cosine bound follows from $-\langle g,d\rangle \ge |g|^2/(1+\tilde q) \ge |g||d|/(1+\tilde q)$. □

Theorem 3.9 (Newton–PCG)**.** *Let d^k be any CG iterate ($j_k \ge 1$) for $H_k d = -g_k$ at v^k, and $\omega_k = \max\{\gamma^i :\ \mathcal{J}_n(v^k+\gamma^i d^k) \le \mathcal{J}_n(v^k) + c\gamma^i(\mathcal{R}_n(v^k),d^k)\}$, $c \in (0,\frac12)$. Then $v^k \in S_0$,*

$$\omega_k \ge \underline{\omega}_N = \min\Big\{1, \frac{2\gamma(1-c)}{1+q_0}\Big\}, \qquad \mathcal{J}_n(v^{k+1}) - \mathcal{J}_n^* \le \Big(1 - \frac{2c\,\underline{\omega}_N}{1+q_0}\Big)\big(\mathcal{J}_n(v^k) - \mathcal{J}_n^*\big).$$

Proof. Note $(\mathcal{R}_n(v),d) = \langle g,d\rangle < 0$. As in Theorem 3.2, for ω with $v^k + sd^k \in S_0$ on $[0,\omega]$, (2.21) and $|d^k|^2 \le -\langle g_k,d^k\rangle$ give

$$\mathcal{J}_n(v^k+\omega d^k) \le \mathcal{J}_n(v^k) + \omega\langle g_k,d^k\rangle + \tfrac{\omega^2}{2}(1+q_0)|d^k|^2 \le \mathcal{J}_n(v^k) + \omega\langle g_k,d^k\rangle\big(1 - \tfrac{\omega(1+q_0)}{2}\big),$$

so the Armijo condition holds for $\omega \le \frac{2(1-c)}{1+q_0}$ and $\omega_k \ge \underline{\omega}_N$. Since $\tilde q_k \le q_0$, (3.12) and (2.15) give $\mathcal{J}_n(v^k) - \mathcal{J}_n(v^{k+1}) \ge c\,\omega_k|g_k|^2/(1+q_0) \ge \frac{2c\underline{\omega}_N}{1+q_0}(\mathcal{J}_n(v^k) - \mathcal{J}_n^*)$. □

The rate depends on N only through θ_N (bounded uniformly by (2.9)) and M_0; local quadratic convergence when j_k grows suitably is standard [21]. In computations the CG relative tolerance is 10^{-2}.

3.5. **A two-sided a posteriori bound.**

Theorem 3.10. *For every $v \in \mathcal{V}_n$, with $M_v = \max\{\|v\|_\infty, \|\phi^{n+1}\|_\infty, \|\phi^n\|_\infty\}$,*

$$\|v-\phi^{n+1}\|_{\mathcal{B}^{-1}} \le \|\mathcal{R}_n(v)\|_{\mathcal{B}} \le \big(1+q(M_v)\big)\|v-\phi^{n+1}\|_{\mathcal{B}^{-1}}. \tag{3.13}$$

Consequently

$$\|\xi\| \le \theta_N^{1/2}\|\mathcal{R}_n(v)\|_{\mathcal{B}}, \quad \|\nabla_N\xi\| \le \tfrac{2}{\sqrt{3}\varepsilon}\|\mathcal{R}_n(v)\|_{\mathcal{B}}, \quad \|\xi\|_\infty \le \tfrac{2C_N}{\sqrt{3}\varepsilon}\|\mathcal{R}_n(v)\|_{\mathcal{B}}, \qquad \xi = v-\phi^{n+1}. \tag{3.14}$$

Proof. With $\xi = v - \phi^{n+1} \in \mathring{\mathcal{V}}$ and $D = D(v,\phi^{n+1};\phi^n) \ge 0$, $\mathcal{R}_n(v) = \mathcal{B}^{-1}\xi + \mathsf{P}_0 D\xi$. Then $\|\xi\|_{\mathcal{B}^{-1}}^2 \le \|\xi\|_{\mathcal{B}^{-1}}^2 + (D\xi,\xi) = (\mathcal{R}_n(v),\xi) \le \|\mathcal{R}_n(v)\|_{\mathcal{B}}\|\xi\|_{\mathcal{B}^{-1}}$, and $\|\mathcal{R}_n(v)\|_{\mathcal{B}} = \|\mathcal{B}\mathcal{R}_n(v)\|_{\mathcal{B}^{-1}} = \|(I+\mathcal{B}\mathsf{P}_0 D)\xi\|_{\mathcal{B}^{-1}} \le (1+q(M_v))\|\xi\|_{\mathcal{B}^{-1}}$ by Lemma 2.4. (3.14) follows from (2.10) and Lemma 2.1(c). □

The lower bound holds without any assumption on v,τ,A,ε,N. The factor $q(M_v)$ involves ϕ^{n+1}; a computable factor follows from Lemma 2.3.

Corollary 3.11 (computable effectivity)**.** *For $v \in \mathcal{V}_n$ let $\mathcal{M}(v) = \max\{\|\phi^n\|_\infty,\ |\operatorname{mean}(\phi^n)| + \frac{2C_N}{\varepsilon}(\mathcal{J}_n(v) + C_{J,n})^{1/2}\}$. Then*

$$\frac{\|\mathcal{R}_n(v)\|_{\mathcal{B}}}{1+q(\mathcal{M}(v))} \le \|v-\phi^{n+1}\|_{\mathcal{B}^{-1}} \le \|\mathcal{R}_n(v)\|_{\mathcal{B}}. \tag{3.15}$$

Proof. $\mathcal{J}_n(\phi^{n+1}) \le \mathcal{J}_n(v)$, so v and ϕ^{n+1} lie in $\{w : \mathcal{J}_n(w) \le \mathcal{J}_n(v)\}$, and (2.17) with $\bar J = \mathcal{J}_n(v)$ gives $M_v \le \mathcal{M}(v)$; q is increasing. □

Algorithm 1: One time step with inexact solve

Input: ϕ^n, ϕ^{n-1}, $\eta \in (0,1)$, $c_R > 0$, an inner method ((3.2), Theorem 3.5 or Theorem 3.9)

$v \leftarrow 2\phi^n - \phi^{n-1}$;
for $k = 0, 1, \dots$ **do**
  $\hat{p} \leftarrow \hat{v} - \widehat{\mathcal{G}_n(v)}$ (one forward transform);
  evaluate $\|\mathcal{R}_n(v)\|$, $(\mathcal{R}_n(v), v - \phi^n)$, $\mathcal{D}_n(v)$ from $\hat{p}, \hat{v}, \hat{\phi}^n, \hat{\phi}^{n-1}$ (Appendix A.6);
  if (3.16) *and* (3.17) *hold* **then return** $\phi^{n+1} = v$;
  one step of the inner method;

Every quantity in $\mathcal{M}(v)$ is evaluated from v, ϕ^n, ϕ^{n-1}. Theorem 3.10 bounds the algebraic error of one step and complements estimators of the discretization error such as [28].

3.6. **Acceptance tests and energy stability of the computed solution.** For $v \in \mathcal{V}_n$ define

$$|(\mathcal{R}_n(v), v - \phi^n)| \le \eta\, \mathcal{D}_n(v), \qquad 0 < \eta < 1, \tag{3.16}$$

$$\|\mathcal{R}_n(v)\| \le c_R \tau^2, \qquad c_R > 0. \tag{3.17}$$

Theorem 3.12. (a) *If $\phi^{n+1} = v$ satisfies (T_E), then*

$$(1-\eta)\mathcal{D}_n(\phi^{n+1}) \le \widetilde{E}(\phi^n, \phi^{n-1}) - \widetilde{E}(\phi^{n+1}, \phi^n) \le (1+\eta)\mathcal{D}_n(\phi^{n+1}). \tag{3.18}$$

If steps $0, \dots, \ell - 1$ satisfy (T_E), then for every $\tau > 0$, $A \ge 0$,

$$E_N(\phi^\ell) \le \widetilde{E}(\phi^\ell, \phi^{\ell-1}) \le E_N(\phi^0), \qquad \|\nabla_N \phi^\ell\|^2 \le \tfrac{2}{\varepsilon^2}\big(E_N(\phi^0) + \tfrac{|\Omega|}{4}\big),$$
$$\sum_{n<\ell} \mathcal{D}_n(\phi^{n+1}) \le (1-\eta)^{-1}\big(E_N(\phi^0) + \tfrac{|\Omega|}{4}\big).$$

(b) *If $v^k \to \phi^{n+1}$ and $v^0 = 2\phi^n - \phi^{n-1}$, Algorithm 1 terminates.*

Proof. (a) By (2.23), $\widetilde{E}(\phi^n, \phi^{n-1}) - \widetilde{E}(\phi^{n+1}, \phi^n) = \mathcal{D}_n - (\mathcal{R}_n, \phi^{n+1} - \phi^n) \in [(1-\eta)\mathcal{D}_n, (1+\eta)\mathcal{D}_n]$. Summing, $\widetilde{E}(\phi^\ell, \phi^{\ell-1}) + (1-\eta)\sum \mathcal{D}_n \le \widetilde{E}(\phi^0, \phi^{-1}) = E_N(\phi^0)$, and $\widetilde{E}(a,b) \ge E_N(a) \ge \frac{\varepsilon^2}{2}\|\nabla_N a\|^2 - \frac{|\Omega|}{4}$ because $\frac{1}{4}s^4 - \frac{1}{2}s^2 \ge -\frac{1}{4}$. (b) $\mathcal{R}_n(v^k) \to 0$, so (T_A) holds eventually. If $\mathcal{D}_n(\phi^{n+1}) > 0$, then $(\mathcal{R}_n(v^k), v^k - \phi^n) \to 0 < \eta\mathcal{D}_n(\phi^{n+1}) = \lim \eta\mathcal{D}_n(v^k)$. If $\mathcal{D}_n(\phi^{n+1}) = 0$, Corollary 2.6 gives $\phi^{n-1} = \phi^n = \phi^{n+1} = v^0$ and both tests hold at $k = 0$. □

Remark 3.13. (3.18) with only the lower inequality needs only $(\mathcal{R}_n(v), v - \phi^n) \le \eta\mathcal{D}_n(v)$. The two-sided ($T_E$) also excludes artificial over-dissipation, $\widetilde{E}(\phi^n, \phi^{n-1}) - \widetilde{E}(\phi^{n+1}, \phi^n) \le (1+\eta)\mathcal{D}_n$; all experiments use it.

3.7. **A fixed number of sweeps.**

Proposition 3.14. *For (3.2) with $\omega_k = \omega$ and $\xi^k = v^k - \phi^{n+1}$,*

$$\xi^K = (1-\omega)^K \xi^0 - \omega \sum_{k=0}^{K-1} (1-\omega)^{K-1-k} T_k \xi^k, \qquad \mathcal{R}_n(v^K) = \mathcal{B}^{-1}(I + T_K)\xi^K, \tag{3.19}$$

with $T_k = \mathcal{B}\mathsf{P}_0 D(v^k, \phi^{n+1}; \phi^n)$. *On a Fourier mode* $k \neq 0$,

$$\beta_k^{-1} = \frac{1}{\tau s_k}\big(1 + \tau A s_k + \tfrac{3}{4}\tau\varepsilon^2 s_k^2\big). \tag{3.20}$$

Proof. (3.11) is $\xi^{k+1} = (1-\omega)\xi^k - \omega T_k \xi^k$; induction gives (3.19). The residual identity is the one in the proof of Theorem 3.10. □

For a smooth solution, $\xi^0 = \tau^2 w + O(\tau^3)$ with w smooth, and for smooth f, (3.20) gives $\mathcal{B}^{-1} f = \tau^{-1}(-\Delta_N)^{-1} f + O(1)$ and $T_k f = \tau(-\Delta_N)\mathsf{P}_0 D_k f + O(\tau^2)$. Inserting into (3.19) formally,

$$\mathcal{R}_n(v^K) = (1-\omega)^K \, \tau \, (-\Delta_N)^{-1} w + O(\tau^2). \tag{3.21}$$

Hence $\|\mathcal{R}_n(v^K)\| = O(\tau)$ for $\omega \neq 1$, while $\|\mathcal{R}_n(v^K)\| = O(\tau^2)$ for $\omega = 1$, $K \geq 1$. By (2.12) the accepted iterate solves the scheme with the consistency defect $-\Delta_N \mathcal{R}_n$; Corollary 4.5 turns $\|\rho^n\| = O(\tau)$ into an $O(\tau)$ error *bound*. For the linearisation at $\phi = 0$ the loss of order is proved.

Proposition 3.15 (linearised fixed sweeps)**.** *Fix* $s > 0$, $A \geq 0$, $\varepsilon > 0$, $K \geq 1$, $\omega \in (0, 1]$, *and put* $r = (1-\omega)^K$, $\lambda = s(1-\varepsilon^2 s)$. *For the scheme linearised at* $\phi = 0$ *on the Fourier mode with symbol* s, *computed with exactly* K *sweeps of* (3.2) *from* $2a_n - a_{n-1}$, *the amplification factor* $g_{\mathrm{phys}}(\tau)$ *of the branch through* $g = 1$ *satisfies*

$$g_{\mathrm{phys}}(\tau) = 1 + \lambda\tau + \Big(\tfrac{1}{2} - \tfrac{r}{1-r}\Big)\lambda^2\tau^2 + O(\tau^3), \qquad \tau^{-1}\log g_{\mathrm{phys}}(\tau) = \lambda - \tfrac{r}{1-r}\lambda^2\tau + O(\tau^2), \tag{3.22}$$

and the other branch tends to r. *Hence* $g_{\mathrm{phys}}^{T/\tau} = e^{\lambda T}\big(1 - \frac{r}{1-r}\lambda^2 T\tau + O(\tau^2)\big)$. *For* $\lambda \neq 0$ *and* $0 < r < 1$ *the physical branch has a nonzero first-order growth-rate error; it vanishes for* $r = 0$ *(*$\omega = 1$*), and for the neutral mode* $\lambda = 0$ *(*$s = \varepsilon^{-2}$*) one has* $g_{\mathrm{phys}} \equiv 1$ *for every* r.

The proof is in Appendix A.5. The statement is modewise for fixed $s, A, \varepsilon, K, \omega$ as $\tau \to 0$; it is not uniform in s or as $r \to 1$. The reduced orders observed in Section 5.2 are consistent with (3.21) and (3.22).

4. Error analysis of the computed solution

Assumption 4.1. The solution u of (1.1) is L-periodic in space and $u \in C^3([0,T]; H^{m+4}_{\mathrm{per}}(\Omega))$ for some $m \geq 2$.

Let P_N be the L^2-projection onto trigonometric polynomials with frequencies $|k_1|, |k_2| < \frac{N}{2}$, U^n the grid values of $\mathsf{P}_N u(t_n)$, $U^{-1} = U^0$, $\bar{t}_n = t_{n+1/2}$, and $M_u = \sup_N \sup_{t \leq T} \|\mathsf{P}_N u(t)\|_{L^\infty} < \infty$. The computed solution starts from $\phi^0 = \phi^{-1} = U^0$ and each step is accepted with residual

$$\rho^n = \mathcal{R}_n(\phi^{n+1}), \qquad e^n = U^n - \phi^n \in \mathring{\mathcal{V}}, \qquad \Xi_n = \Big(\tau \sum_{j=0}^{n-1} \|\rho^j\|^2\Big)^{1/2}.$$

Ξ_n is computable: $\|\rho^j\|$ is evaluated in Algorithm 1.

Lemma 4.2 (consistency)**.** *Let* $\zeta^n = \tau^{-1}(U^{n+1} - U^n) - \Delta_N \mu_n(U^{n+1})$, *with* μ_n *formed from* (U^n, U^{n-1}). *Then* $\zeta^n \in \mathring{\mathcal{V}}$ *and, under Assumption 4.1,*

$$\zeta^n = \zeta_1^n - \tau^2 \Delta \mathsf{P}_N W(\bar{t}_n) + r^n, \quad \|\zeta_1^n\|_{-1} \leq \tfrac{\tau^2}{24}\sup_t \|u_{ttt}\|_{-1}, \quad \|r^n\|_{-1} \leq C(\tau^3 + h^m) \ (n \geq 1), \tag{4.1}$$

$$W = \big(\tfrac{3}{8} + A\big)u_{tt} - \tfrac{3}{8}\varepsilon^2 \Delta u_{tt} + \tfrac{3}{8}u^2 u_{tt} + \tfrac{1}{4}u\,u_t^2, \qquad \|\zeta^0\|_{-1} \leq C(\tau + h^m),$$

where $\zeta_1^n = \mathsf{P}_N[\tau^{-1}(u(t_{n+1}) - u(t_n)) - u_t(\bar{t}_n)]$. *In particular* $\|\zeta^n\|_{-1} \leq C_u(\tau^2 + h^m)$ *for* $n \geq 1$.

The proof is in Appendix A.4.

Theorem 4.3. *Let Assumption 4.1 hold and $A \ge 0$. There are $\tau_0, c_0, C > 0$ depending on u, ε, A, L, T only such that, if $\tau \le \tau_0$ and*

$$(1+\ln N)^{1/2}\,\tau^{-1/2}\big(\tau^2 + h^m + \Xi_n\big) \le c_0 \qquad \text{for all } t_n \le T, \tag{4.2}$$

then for all $t_n \le T$

$$\|e^n\|_{-1} + \Big(\tau \sum_{j=1}^{n} \varepsilon^2 \|\nabla_N e^j\|^2\Big)^{1/2} \le C\big(\tau^2 + h^m + \Xi_n\big), \qquad \|\phi^n\|_\infty \le M_u + 1. \tag{4.3}$$

Proof. Let $\hat M = M_u + 1$, $\hat\lambda = \frac{3}{2}\hat M^2$,

$$K_2 = \tfrac{5}{4}, \qquad K_3 = \tfrac{1}{2}\big(\hat\lambda + \tfrac{3}{2}\big)^2 + 4A, \qquad K_4 = \tfrac{1}{4} + A, \qquad C_\varepsilon = \tfrac{1}{2} + \tfrac{2}{\varepsilon^2}\max\{K_2, K_3, K_4\}^2, \tag{4.4}$$

and $a_j = \|\nabla_N e^j\|^2$, $b_j = \|e^j\|_{-1}^2$. We prove by induction on $k \ge 1$: if

$$\|\phi^j\|_\infty \le \hat M \qquad (0 \le j \le k-1), \tag{4.5}$$

then (4.3) holds for $n = k$. (4.5) holds for $k = 1$.

Error equation. Subtracting (2.12) with $v = \phi^{n+1}$ from the definition of ζ^n (here $\mu_n(U^{n+1})$ is built from (U^n, U^{n-1}) and $\mu_n(\phi^{n+1})$ from (ϕ^n, ϕ^{n-1})),

$$\tau^{-1}(e^{n+1} - e^n) = \Delta_N\big(\mu_n(U^{n+1}) - \mu_n(\phi^{n+1})\big) + \zeta^n + \Delta_N \rho^n. \tag{4.6}$$

By symmetry of χ and (2.18),

$$\chi(U^{n+1}, U^n) - \chi(\phi^{n+1}, \phi^n) = \underbrace{D(U^{n+1}, \phi^{n+1}; \phi^n)}_{\mathfrak{p}_n \ge 0} e^{n+1} + \underbrace{D(U^n, \phi^n; U^{n+1})}_{\mathfrak{q}_n} e^n, \tag{4.7}$$

and $0 \le \mathfrak{q}_n \le \hat\lambda$ for $n \le k-1$ by (2.19), (4.5). Hence

$$\begin{aligned} \mu_n(U^{n+1}) - \mu_n(\phi^{n+1}) = \mathfrak{p}_n e^{n+1} + \mathfrak{q}_n e^n - \big(\tfrac{3}{2}e^n - \tfrac{1}{2}e^{n-1}\big) \\ - \varepsilon^2 \Delta_N\big(\tfrac{3}{4}e^{n+1} + \tfrac{1}{4}e^{n-1}\big) + A\big(e^{n+1} - 2e^n + e^{n-1}\big). \end{aligned} \tag{4.8}$$

Step $n = 0$. $e^0 = e^{-1} = 0$ and $\mathfrak{q}_0 e^0 = 0$. Testing (4.6) with $\tau(-\Delta_N)^{-1}e^1$,

$$b_1 + \tau(\mathfrak{p}_0 e^1, e^1) + \tfrac{3}{4}\tau\varepsilon^2 a_1 + \tau A\|e^1\|^2 = \tau(\zeta^0, (-\Delta_N)^{-1}e^1) - \tau(\rho^0, e^1) \le \tau\|\zeta^0\|_{-1} b_1^{1/2} + \tfrac{\tau L}{2\pi}\|\rho^0\| a_1^{1/2},$$

by Lemma 2.1(b). With $xy \le \frac{1}{2}x^2 + \frac{1}{2}y^2$ and $xy \le \frac{\varepsilon^2}{4}y^2 + \frac{1}{\varepsilon^2}x^2$,

$$b_1 + \tau\varepsilon^2 a_1 \le \tau^2\|\zeta^0\|_{-1}^2 + \tfrac{L^2}{2\pi^2\varepsilon^2}\tau\|\rho^0\|^2 \le C\big(\tau^4 + h^{2m} + \tau\|\rho^0\|^2\big). \tag{4.9}$$

Step $1 \le n \le k-1$. Testing (4.6) with $\tau(-\Delta_N)^{-1}e^{n+1}$ and using $(x-y, x) = \frac{1}{2}(\|x\|^2 - \|y\|^2 + \|x-y\|^2)$,

$$\begin{aligned} \tfrac{1}{2}\big(b_{n+1} - b_n + \|e^{n+1} - e^n\|_{-1}^2\big) + \tau(\mu_n(U^{n+1}) - \mu_n(\phi^{n+1}), e^{n+1}) \\ = \tau(\zeta^n, (-\Delta_N)^{-1}e^{n+1}) - \tau(\rho^n, e^{n+1}). \end{aligned}$$

Writing $x_j = \|e^j\|$, the terms of (4.8) satisfy

$$\begin{aligned} (\mathfrak{p}_n e^{n+1} + \mathfrak{q}_n e^n, e^{n+1}) &\ge -\hat\lambda x_n x_{n+1}, \\ -(\tfrac{3}{2}e^n - \tfrac{1}{2}e^{n-1}, e^{n+1}) &\ge -\tfrac{3}{2}x_n x_{n+1} - \tfrac{1}{2}x_{n-1}x_{n+1}, \\ \varepsilon^2\big\langle \tfrac{3}{4}e^{n+1} + \tfrac{1}{4}e^{n-1}, e^{n+1}\big\rangle_1 &\ge \tfrac{3}{4}\varepsilon^2 a_{n+1} - \tfrac{1}{4}\varepsilon^2 a_{n-1}^{1/2} a_{n+1}^{1/2} \ge \tfrac{5}{8}\varepsilon^2 a_{n+1} - \tfrac{1}{8}\varepsilon^2 a_{n-1}, \\ A(e^{n+1} - (2e^n - e^{n-1}), e^{n+1}) &\ge \tfrac{A}{2}x_{n+1}^2 - \tfrac{A}{2}\|2e^n - e^{n-1}\|^2 \ge -4Ax_n^2 - Ax_{n-1}^2, \end{aligned}$$

and the right-hand side is at most $\frac{\tau}{2}(\|\zeta^n\|_{-1}^2+b_{n+1})+\frac{\tau}{2}(\|\rho^n\|^2+x_{n+1}^2)$. With $(\hat\lambda+\frac32)x_nx_{n+1}\le \frac12x_{n+1}^2+\frac12(\hat\lambda+\frac32)^2x_n^2$ and $\frac12x_{n-1}x_{n+1}\le\frac14x_{n+1}^2+\frac14x_{n-1}^2$,

$$\tfrac12(b_{n+1}-b_n)+\tau\varepsilon^2\big(\tfrac58a_{n+1}-\tfrac18a_{n-1}\big)\le\tfrac{\tau}{2}\big(\|\zeta^n\|_{-1}^2+\|\rho^n\|^2+b_{n+1}\big)+\tau\big(K_2x_{n+1}^2+K_3x_n^2+K_4x_{n-1}^2\big). \tag{4.10}$$

By Lemma 2.1(a), $Kx_j^2\le Kb_j^{1/2}a_j^{1/2}\le\frac{\varepsilon^2}{8}a_j+\frac{2K^2}{\varepsilon^2}b_j$. Hence

$$\tfrac12(b_{n+1}-b_n)+\tau\varepsilon^2\big(\tfrac12a_{n+1}-\tfrac18a_n-\tfrac14a_{n-1}\big)\le\tfrac{\tau}{2}\big(\|\zeta^n\|_{-1}^2+\|\rho^n\|^2\big)+\tau C_\varepsilon\big(b_{n+1}+b_n+b_{n-1}\big). \tag{4.11}$$

Summation. For $2\le k'\le k$, summing (4.11) over $1\le n\le k'-1$ with $a_0=0$,

$$\sum_{n=1}^{k'-1}\big(\tfrac12a_{n+1}-\tfrac18a_n-\tfrac14a_{n-1}\big)\ge\big(\tfrac12-\tfrac18-\tfrac14\big)\sum_{j=2}^{k'}a_j-\big(\tfrac18+\tfrac14\big)a_1=\tfrac18\sum_{j=2}^{k'}a_j-\tfrac38a_1,$$

so that, with $\Phi_{k'}=\frac12b_1+\frac38\tau\varepsilon^2a_1+\frac{\tau}{2}\sum_{n=1}^{k'-1}(\|\zeta^n\|_{-1}^2+\|\rho^n\|^2)$,

$$\tfrac12b_{k'}+\tfrac18\tau\varepsilon^2\sum_{j=2}^{k'}a_j\le\Phi_{k'}+3\tau C_\varepsilon\sum_{j=0}^{k'}b_j. \tag{4.12}$$

By (4.9), Lemma 4.2 and $\tau\|\zeta^0\|_{-1}\le C(\tau^2+\tau h^m)$, $\Phi_{k'}\le\Phi:=C_\Phi^2(\tau^2+h^m+\Xi_k)^2$ with C_Φ independent of k,τ,N.

Gronwall. Let $\tau_0\le(12C_\varepsilon)^{-1}$. From (4.12), $\frac14b_{k'}\le\Phi+3\tau C_\varepsilon\sum_{j<k'}b_j$ for $2\le k'\le k$; this also holds for $k'=0,1$ since $b_0=0$ and $b_1\le2\Phi$ by (4.9). Lemma A.1 gives $b_{k'}\le4\Phi e^{12C_\varepsilon t_{k'}}$. Inserting into (4.12) with $k'=k$ and adding (4.9),

$$b_k+\tau\varepsilon^2\sum_{j=1}^{k}a_j\le C_*^2\big(\tau^2+h^m+\Xi_k\big)^2,\qquad C_*^2=C\,C_\Phi^2\big(1+TC_\varepsilon\big)e^{12C_\varepsilon T}, \tag{4.13}$$

which is (4.3) for $n=k$.

Closing the induction. From (4.13), $a_k^{1/2}\le C_*\varepsilon^{-1}\tau^{-1/2}(\tau^2+h^m+\Xi_k)$, and by Lemma 2.1(c)

$$\|\phi^k\|_\infty\le\|U^k\|_\infty+\|e^k\|_\infty\le M_u+\tfrac{\sqrt2}{\pi}(1+\ln N)^{1/2}C_*\varepsilon^{-1}\tau^{-1/2}\big(\tau^2+h^m+\Xi_k\big)\le M_u+1$$

if $c_0\le\pi\varepsilon/(\sqrt2C_*)$. So (4.5) holds for $k+1$. □

Corollary 4.4 (accuracy test)**.** *If every step satisfies (T_A), then $\Xi_n\le c_R\sqrt{T}\tau^2$ and, under $\tau\le\tau_0$, $(1+\ln N)^{1/2}(\tau^{3/2}+\tau^{-1/2}h^m)\le c_0'$,*

$$\|e^n\|_{-1}+\Big(\tau\sum_{j=1}^{n}\varepsilon^2\|\nabla_Ne^j\|^2\Big)^{1/2}\le C(\tau^2+h^m).$$

Exact solves are the case $c_R=0$. With (T_E) also imposed, Theorem 3.12 holds for the same sequence.

Corollary 4.5 (residual of order τ)**.** *If $\|\rho^n\|\le c_1\tau$ for all n, then $\Xi_n\le c_1\sqrt{T}\tau$, and under $(1+\ln N)^{1/2}(\tau^{1/2}+\tau^{-1/2}h^m)\le c_0'$ the error is $O(\tau+h^m)$. By (3.21) this is the regime of a fixed number of relaxed sweeps.*

Remark 4.6. (i) (4.2) is used only for $\|\phi^n\|_\infty\le\hat M$. Truncating the quartic potential to one with globally Lipschitz derivative [8] removes it, as in the fully discrete analysis [28], but changes the scheme; for the quartic potential the maximum norm must be controlled. In three dimensions $(1+\ln N)^{1/2}$ becomes $CN^{1/2}$. (ii) $C_*\sim e^{CT\varepsilon^{-2}}$ through C_ε; polynomial dependence

TABLE 1. Accuracy test, all inner-solve variants: observed orders p_j, error at $\tau_{10} = 2^{-10}$ relative to the exact solve, and range over τ of the mean number of inner iterations per step.

	$p_j = \log_2(e_{j-1}/e_j)$, $\tau_j = 2^{-j}$						$e_{10}/e_{10}^{\mathrm{ex}}$	mean it.
inner solve	$j=5$	$j=6$	$j=7$	$j=8$	$j=9$	$j=10$		
Newton, $\|\mathcal{R}_n\|_{\mathcal{B}} \le 10^{-12}$ (exact)	1.95	1.96	1.99	2.00	2.00	2.00	1.00	2.65–4.19
$\omega = 1$, $K = 1$	1.88	1.97	1.99	2.00	2.00	2.00	0.84	1.00–1.00
$\omega = 1$, $K = 2$	2.08	1.96	2.00	2.01	2.01	2.01	1.00	2.00–2.00
$\omega = 1$, $K = 6$ (original code)	1.95	1.96	1.99	2.00	2.00	2.00	1.00	6.00–6.00
$\omega = \frac{1}{2}$, $K = 1$	1.77	1.79	1.69	1.56	1.39	1.25	9.68	1.00–1.00
$\omega = \frac{1}{2}$, $K = 2$	1.96	1.86	1.83	1.72	1.57	1.40	3.86	2.00–2.00
$\omega = \frac{1}{2}$, $K = 4$	1.95	1.94	1.95	1.92	1.85	1.73	1.56	4.00–4.00
$\omega = \frac{1}{2}$ + (T_A), $c_R = 1$	1.93	1.79	1.70	2.12	2.00	2.21	2.21	0.88–3.00
Armijo + (T_A), $c_R = 1$	1.82	2.31	2.07	2.80	2.00	2.00	0.84	0.62–1.00
Armijo + (T_A), $c_R = 0.1$	2.14	1.98	2.05	2.05	2.02	2.02	0.88	1.94–3.03

on ε^{-1} requires a spectral estimate for the linearized operator [5, 1, 14]. (iii) (4.3) separates the discretization error $\tau^2 + h^m$ from the algebraic error Ξ_n, which is computable.

5. Numerical experiments

5.1. **Setting.** Real 2D FFTs in double precision; work is the number of forward plus inverse transforms, including conjugate-gradient work for Newton. Unless stated otherwise: $v^0 = 2\phi^n - \phi^{n-1}$; Armijo with $c = \frac{1}{4}$, $\gamma = \frac{1}{2}$, $\hat{\omega}_k = \min\{1, 2\omega_{k-1}\}$; Newton–PCG (Theorem 3.9) with CG relative tolerance 10^{-2}; *practical* rule: Algorithm 1 with $\eta = \frac{1}{2}$, $c_R = 1$ and the two-sided (T_E); *tight* rule: $\|\mathcal{R}_n\|_{\mathcal{B}} \le 10^{-10}$. A step is aborted (failure) if it exceeds its iteration cap (300 for gradient methods, 60 for Newton unless stated) or $\|\mathcal{R}_n\|_{\mathcal{B}} > 10^8$; a run diverges if an accepted iterate is non-finite or exceeds 10^3 in $\|\cdot\|_\infty$. Norms: $\|\cdot\|$ discrete L^2; $\|\cdot\|_{-1}$; $\|\nabla_N \cdot\|$ the H^1 *seminorm*; $\|\cdot\|_\infty$. Random data are uniform in $[-0.1, 0.1]$ (NumPy default generator, stated seed) or band-limited (Section 5.7). Simulation, diagnostics and plotting are separate scripts; every number below is extracted from saved data. In Figs. 2 and 3: Newton–PCG blue circles, Armijo green squares, relaxed $\omega = \frac{1}{2}$ orange triangles, linear $\omega = 1$ purple diamonds; solid residual stopping, dashed fixed K; filled practical, open tight.

5.2. **Accuracy.** Manufactured solution $\phi = \sin(2\pi x)\cos(2\pi y)\cos t$ on $(0,1)^2$ with forcing at $t_{n+1/2}$ (a linear term in $\mathcal{J}_n$, a constant in $\mathcal{R}_n$, a forcing term in (2.23)), ϕ^{-1} exact, $\varepsilon = 0.1$, $A = 1$, $N = 48$, $T = 1$, $\tau = 2^{-4}, \dots, 2^{-10}$; one wavenumber shell, and ϕ^3 is resolved exactly at $N = 48$, so the error is temporal. Error $e_j = \max_n \|e^n\|_{-1}$ at $\tau_j = 2^{-j}$.

Figure 1 and Table 1 are consistent with (3.21) and Proposition 3.15:

- $\omega = \frac{1}{2}$, fixed $K = 1, 2, 4$: p_j decreases to $1.25, 1.40, 1.73$ at $j = 10$; $e_{10}/e_{10}^{\mathrm{ex}} = 9.7, 3.9, 1.6$.
- $\omega = 1$, fixed K: $p_j \in [1.88, 2.08]$; $e_{10}/e_{10}^{\mathrm{ex}} = 0.84, 1.00, 1.00$ for $K = 1, 2, 6$ (leading term of (3.21) vanishes).
- $\omega = \frac{1}{2}$ + (T_A): $p_j \in [1.70, 2.21]$, $e_{10}/e_{10}^{\mathrm{ex}} = 2.21$, 0.88–3.00 iterations. Armijo + (T_A): $c_R = 1$, $p_j \in [1.82, 2.80]$; $c_R = 0.1$, $p_j \in [1.98, 2.14]$, ratio 0.88–1.12 over τ.
- The isolated $p_8 = 2.80$ (Armijo, $c_R = 1$) coincides with the change of the mean iteration count from 0.97 to 1.00; we attribute it to the switch between accepting the predictor and one iteration.

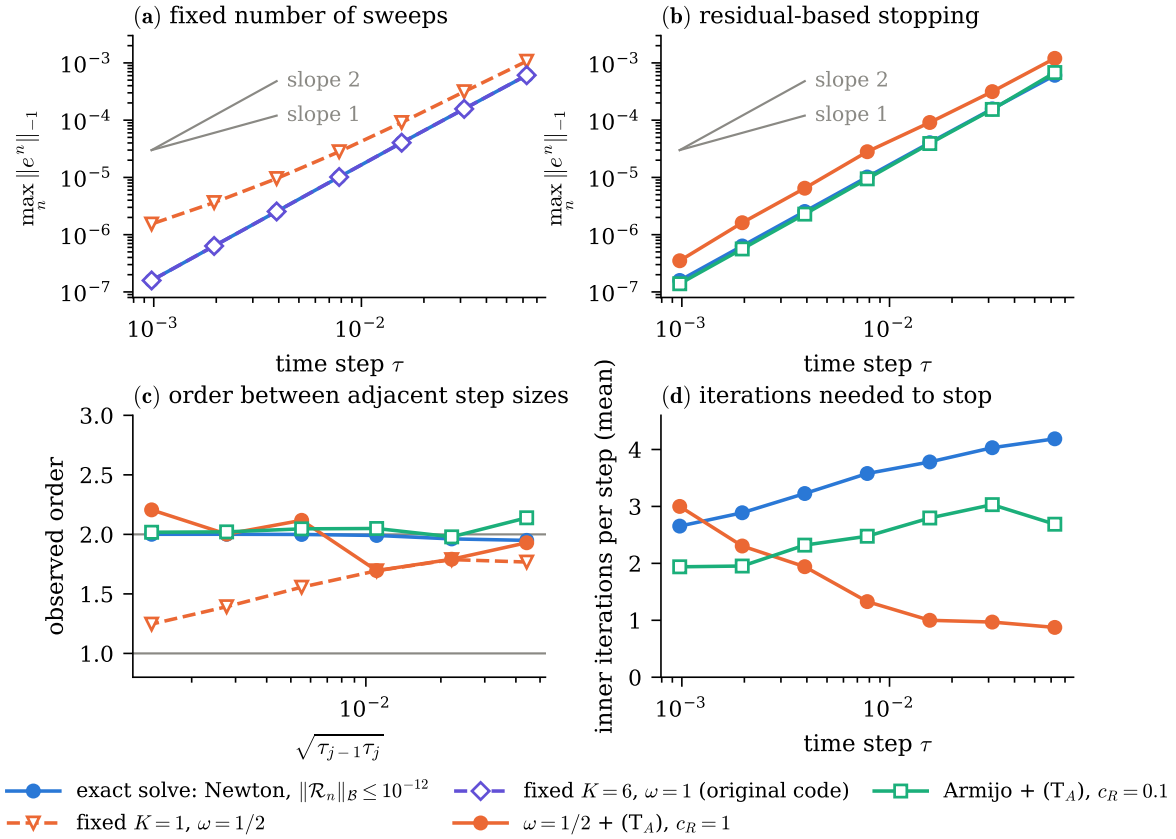


FIGURE 1. Accuracy (manufactured solution, $(0,1)^2$, $N = 48$, $\varepsilon = 0.1$, $A = 1$, $T = 1$). (a,b) $\max_n \|e^n\|_{-1}$ against τ, fixed sweep counts (a) and residual-based stopping (b); exact solve (Newton, $\|\mathcal{R}_n\|_{\mathcal{B}} \le 10^{-12}$) in both. (c) $\log(e_{j-1}/e_j)/\log(\tau_{j-1}/\tau_j)$ at $\sqrt{\tau_{j-1}\tau_j}$. (d) Mean inner iterations per step; values below one mean that the predictor passes (T_A) in some steps. All ten variants are in Table 1.

Spatial accuracy ($\phi = 0.4\cos t \sin(2\pi x)e^{\cos(2\pi y)}$, $\tau = 10^{-4}$, $T = 0.01$, exact solves): discrete L^2 errors $4.0\cdot10^{-3}$, $2.6\cdot10^{-5}$, $2.4\cdot10^{-7}$, $3.5\cdot10^{-9}$ for $N = 8, 12, 16, 20$, and $2.8\cdot10^{-9}$ for $24 \le N \le 64$ (an N-independent floor, attributed to the time discretization).

5.3. **Robustness and cost.** *One hard step* (Fig. 2(a,b)): step $n = 20$ ($t = 0.08$, end of the spinodal stage, $\max|\phi^n| \approx 1.05$) of the problem $[0, 6.4)^2$, $N = 256$, $\varepsilon = 0.02$, $\tau = 0.004$, seed 1; common history from Newton; runs to $\|\mathcal{R}_n\|_{\mathcal{B}} \le 10^{-12}$ or 150 iterations. Transforms to acceptance (practical / tight 10^{-10}):

	linear $\omega = 1$	relaxed $\omega = \frac{1}{2}$	Armijo	Newton–PCG
$A = 0$, $\theta_N = 1.83$	diverges ($\|\mathcal{R}_n\|_{\mathcal{B}} : 0.20 \to 3\cdot10^{17}$)	53 / 149	31 / 69	35 / 59
$A = 2$, $\theta_N = 0.39$	115 / 255	33 / 65	25 / 47	27 / 43

Whole runs (Table 2, $T = 1$): the unrelaxed iteration diverges in 4 of 6 settings. Corollary 3.3 guarantees convergence if $M_0 < (2/(3\theta_N))^{1/2} = 0.85, 0.60, 0.43$ ($A = 0$) and $1.44, 1.30, 1.23$ ($A = 2$) for $\tau = 0.001, 0.004, 0.016$; the divergent runs fail after the amplitude exceeds these

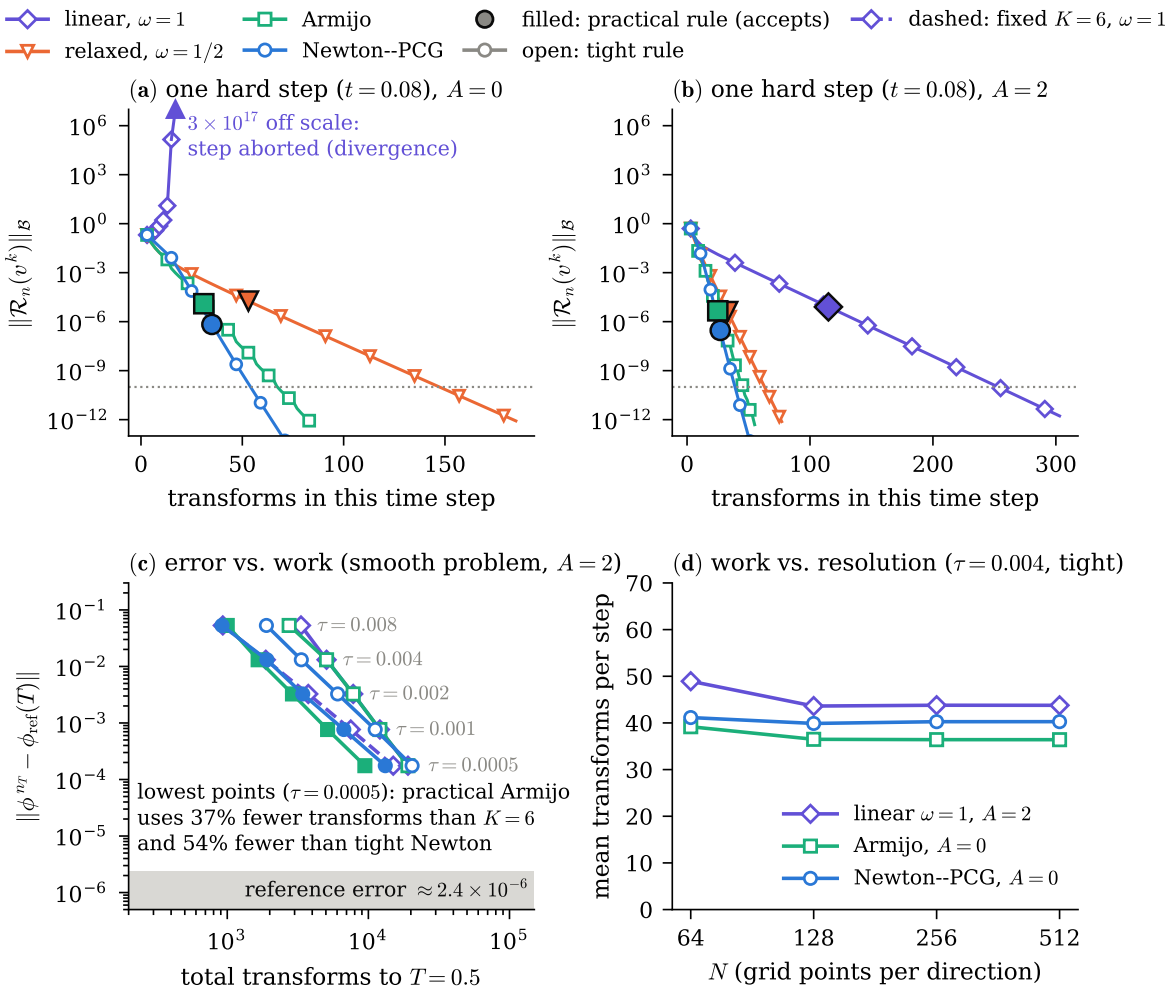


FIGURE 2. Robustness and cost. (a,b) $\|\mathcal{R}_n(v^k)\|_{\mathcal{B}}$ against cumulative transforms in step $n = 20$ ($t = 0.08$) of $[0, 6.4)^2$, $N = 256$, $\varepsilon = 0.02$, $\tau = 0.004$, seed 1; large filled markers: practical acceptance; dotted line: 10^{-10}. In (a) the last residual of the unrelaxed iteration, $3 \cdot 10^{17}$, lies above the range (triangle); the step is then aborted. (c) Discrete L^2 error at $T = 0.5$ against total transforms for the smooth interface problem ($[0, 2)^2$, $N = 128$, $\varepsilon = 0.04$, $A = 2$, $\tau = 0.008, \ldots, 0.0005$), reference $\tau = 6.25 \cdot 10^{-5}$, Newton to 10^{-12}; shaded: below the Richardson estimate $2.4 \cdot 10^{-6}$ of the reference error; annotation: savings at the smallest τ from the unrounded counts (9444 against 15000 and 20412). (d) Mean transforms per step against N ($[0, 6.4)^2$, $\varepsilon = 0.04$, $\tau = 0.004$, $T = 1$, tight rule, modes $|k_i| \le 16$).

values (the condition is sufficient, not necessary). $\omega = \frac{1}{2}$ fails only for $A = 0$, $\tau = 0.016$, where $\frac{1}{2} > 2/(1 + q_0)$ as soon as $M_0 > 0.74$. Armijo and Newton converge in all settings.

TABLE 2. $[0, 6.4)^2$, $N = 256$, $\varepsilon = 0.02$, seed 1, $T = 1$: mean inner iterations (it) and transforms (FFT) per step; "div." divergence. $\theta_N = 0.91, 1.83, 3.65$ ($A = 0$), $0.32, 0.39, 0.44$ ($A = 2$) for $\tau = 0.001, 0.004, 0.016$.

		$\tau = 0.001$				$\tau = 0.004$				$\tau = 0.016$			
		$A = 0$		$A = 2$		$A = 0$		$A = 2$		$A = 0$		$A = 2$	
stop	method	it	FFT	it	FFT	it	FFT	it	FFT	it	FFT	it	FFT
$\|\mathcal{R}_n\|_{\mathcal{B}} \le 10^{-10}$	linear, $\omega = 1$	div.		18.8	41	div.		41.0	85	div.		div.	
	relaxed, $\omega = \frac{1}{2}$	22.4	48	23.1	49	83.7	170	26.8	57	div.		30.2	63
	Armijo	15.0	33	18.8	41	28.1	59	18.6	40	57.2	117	19.8	43
	Newton–PCG	3.4	36	3.2	27	4.4	54	4.0	34	5.5	81	4.8	41
(T_E)+(T_A)	linear, $\omega = 1$	div.		8.2	19	div.		16.0	35	div.		div.	
	relaxed, $\omega = \frac{1}{2}$	11.1	25	11.7	26	21.6	46	11.2	25	div.		10.5	24
	Armijo	7.2	17	8.2	19	10.5	24	8.1	19	17.2	37	7.2	17
	Newton–PCG	2.0	22	2.1	18	2.2	27	2.2	20	2.6	39	2.3	22

Work at measured accuracy (Fig. 2(c)). The random problem is unsuitable: halving τ changes it by $O(1)$ (Section 5.5), reflecting $e^{CT\varepsilon^{-2}}$ in Theorem 4.3. We use, on $[0, 2)^2$ with $L = 2$,

$$\phi^0 = \tanh \frac{\max\{d_\star, d_1, d_2\}}{\sqrt{2}\varepsilon}, \qquad d_\star = 0.22L + 0.04L \cos 6\vartheta - r, \qquad d_i = 0.11L - |x - c_i|,$$

with polar coordinates (r, ϑ) about $(0.32L, 0.5L)$ (a six-lobed star), $c_1 = (0.7L, 0.36L)$, $c_2 = (0.7L, 0.64L)$; the circles merge and the star relaxes; $\|\phi_{\mathrm{ref}}(T)\| = 1.87$. All six variants have the same error at each τ (spread $< 0.05\%$), with order 2.01–2.13, and none fails. Transforms relative to the original code ($\omega = 1$, $K = 6$): practical Armijo $1.08, 0.88, 0.77, 0.69, 0.63$ for $\tau = 0.008, \dots, 0.0005$; practical Newton 0.88–1.02; tight variants 1.27–3.58. Against the same method run to the tight tolerance, the practical rule uses 50.5–67.3% (Armijo) and 35.4–51.5% (Newton) fewer transforms. The original code is as accurate here ($A = 2$, smooth data) but diverges for $A = 0$ (panel (a)).

Mesh (Fig. 2(d)): from $N = 64$ to 512 the transforms per step change $49 \to 44$ (linear, $A = 2$), $39 \to 36$ (Armijo, $A = 0$), $41 \to 40$ (Newton, $A = 0$), consistent with the N-independence of θ_N in Theorems 3.7 and 3.9.

5.4. **Reliability of the a posteriori bound.** $[0, 3.2)^2$, $N = 128$, $\varepsilon = 0.02$, $A = 2$, $\tau = 0.004$, seed 3; reference step solutions by Newton to $\|\mathcal{R}_n\|_{\mathcal{B}} \le 10^{-13}$. (a) 739 iterates (first 15 of $\omega = 1$ and $\omega = \frac{1}{2}$ at $n = 0, 10, \dots, 240$): $\|v - \phi^{n+1}\|_{\mathcal{B}^{-1}} / \|\mathcal{R}_n(v)\|_{\mathcal{B}} \in [0.589, 0.9996]$ ($\omega = 1$: ≥ 0.589; $\omega = \frac{1}{2}$: ≥ 0.654), against the guaranteed interval $[1/(1+q_{\max}), 1] = [0.448, 1]$ of (3.13) with $q_{\max} = 1.23$. (b) First iterate with $\|\mathcal{R}_n\|_{\mathcal{B}} \le \mathrm{tol}$, $\mathrm{tol} = 10^{-2}, \dots, 10^{-10}$, four methods, $n = 5, 15, \dots, 245$: 500 accepted iterates, error/tol $\in [2.2 \cdot 10^{-3}, 0.974]$, no violation.

5.5. **Coarsening benchmark.** The problem of [7]: $[0, 12.8)^2$, $N = 512$, $\varepsilon = 0.02$, $A = 0$, $\tau = 0.004$, $T = 10$, seed 1 (mean $-8.7 \cdot 10^{-5}$); practical and tight solvers in lockstep, i.e. two solvers of the *same* fully discrete problem. The trajectory is not converged in time: the fastest linear mode has rate $\lambda_{\max} = (4\varepsilon^2)^{-1} = 625$, $\tau\lambda_{\max} = 2.5$, and halving τ changes the solution by 8.7 ($t = 0.2$) and 12.2 ($t = 1$) in L^2; the energies $E + |\Omega|/4$ of the τ and $\tau/2$ runs differ by 0.4% and 0.8%.

Practical: 7.3 iterations per step (max 18), 17.6 transforms; tight: 3.6 Newton iterations (max 6), 45.0 transforms; all steps accepted with $\|\rho^n\| \le 1.6 \cdot 10^{-5} = c_R\tau^2$. Differences (Figs. 4 and 5): snapshot median $\le 1.5 \cdot 10^{-6}$, 99th percentile $\le 2.2 \cdot 10^{-4}$, maximum $\le 1.35 \cdot 10^{-2}$; over the run $\max_n \|\phi_{\mathrm{p}}^n - \phi_{\mathrm{t}}^n\| = 4.7 \cdot 10^{-3}$ ($t \approx 2.5$; relative $\le 4 \cdot 10^{-4}$ since $\|\phi\| \in [11.0, 12.4]$),

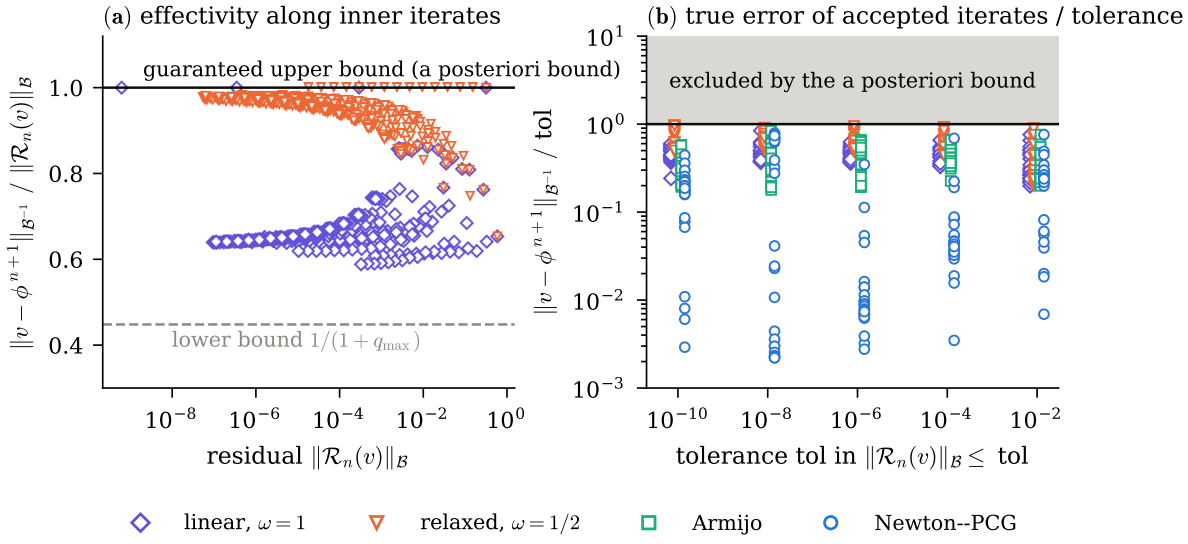


FIGURE 3. A posteriori bound $([0,3.2)^2$, $N=128$, $\varepsilon=0.02$, $A=2$, $\tau=0.004$, seed 3; references by Newton to 10^{-13}). (a) $\|v-\phi^{n+1}\|_{\mathcal{B}^{-1}}/\|\mathcal{R}_n(v)\|_{\mathcal{B}}$ for 739 iterates; dashed: $1/(1+q_{\max})$ with $q_{\max}=\max q(M_v)$, M_v as in Theorem 3.10, $\theta_N=0.39$. (b) True error/tolerance for the first accepted iterate (500 iterates, methods offset horizontally); the shaded region is excluded by Theorem 3.10.

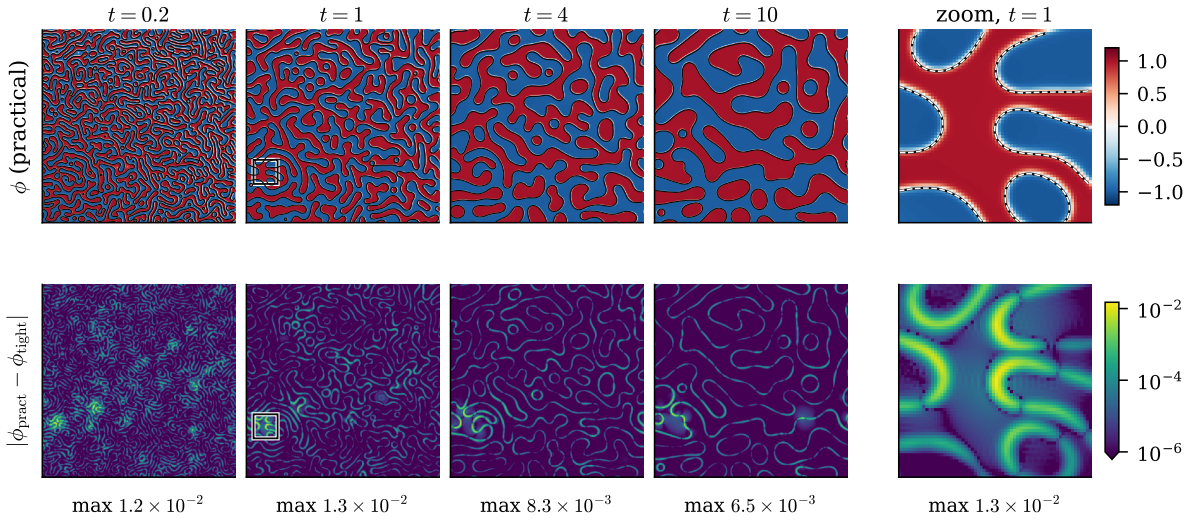


FIGURE 4. Coarsening benchmark $([0,12.8)^2$, $N=512$, $\varepsilon=0.02$, $A=0$, $\tau=0.004$, seed 1); both solvers solve the same discrete problem, which is not converged in τ. Columns 1–4: full domain at $t=0.2,1,4,10$; column 5: the 1.6×1.6 box (64^2 points) marked at $t=1$, centred on the largest difference. Top: practical ϕ, limits $[-1.2,1.2]$ (all values in $[-1.149,1.040]$); black: tight zero level set; white dashed (column 5): practical zero level set. Bottom: $|\phi_{\text{pract}}-\phi_{\text{tight}}|$ on $[10^{-6},1.5\cdot10^{-2}]$ (contains every maximum, stated beneath).

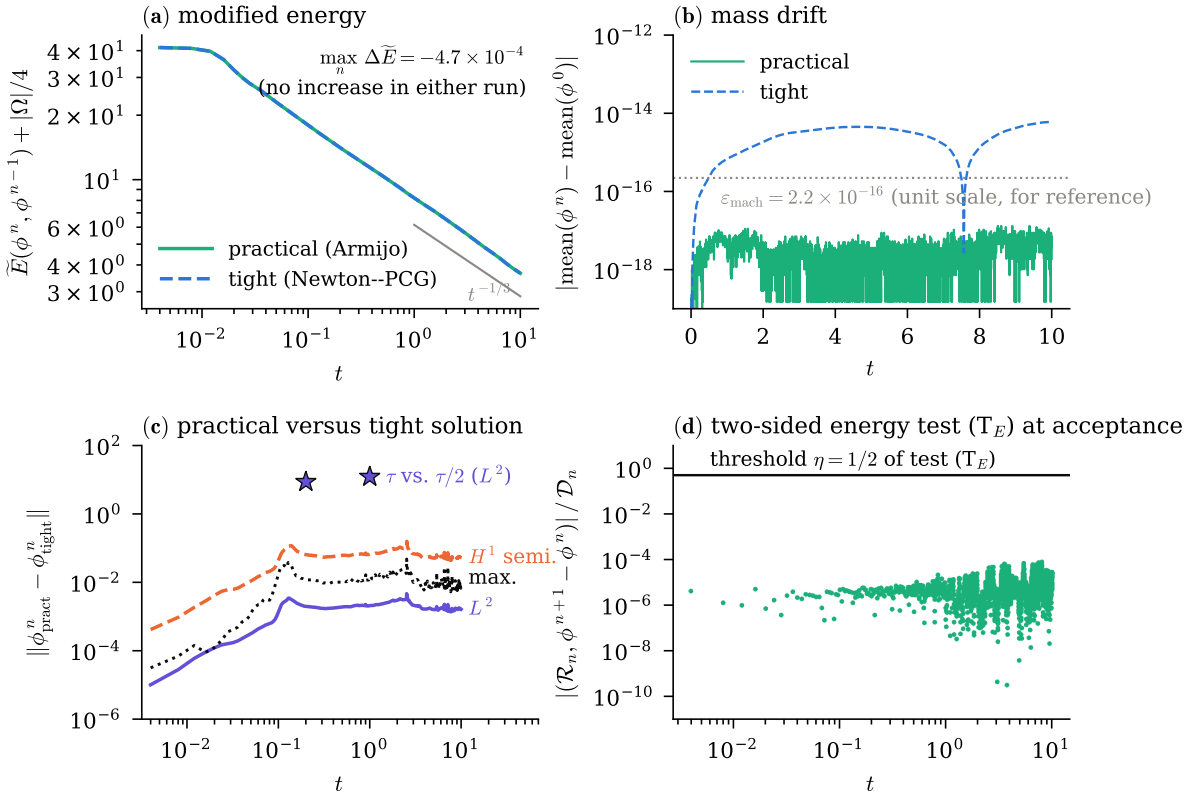


FIGURE 5. Structure on the benchmark of Fig. 4; practical solid (green), tight dashed (blue). (a) $\widetilde{E}(\phi^n, \phi^{n-1}) + |\Omega|/4$; grey: slope $-\frac{1}{3}$. (b) $|\,\mathrm{mean}(\phi^n) - \mathrm{mean}(\phi^0)|$; dotted: machine epsilon $2.2 \cdot 10^{-16}$ as a unit-scale reference, not a bound. (c) Practical minus tight in L^2, H^1 seminorm and maximum norm; stars: L^2 difference of tight solutions with τ and $\tau/2$. (d) $|(\mathcal{R}_n(v), v - \phi^n)|/\mathcal{D}_n(v)$ at acceptance; the two-sided test requires $\leq \eta = \frac{1}{2}$ and is stronger than needed for decay (Remark 3.13).

H^1 seminorm < 0.16, maximum norm < 0.05. $\widetilde{E}$ decreases at every step of both runs (largest change $-4.7 \cdot 10^{-4}$; relative difference of $\widetilde{E} + |\Omega|/4$ between runs $\leq 1.7 \cdot 10^{-5}$); between $t = 1$ and 10, $\widetilde{E} + |\Omega|/4 \sim t^{-0.36}$, close to $t^{-1/3}$ [19]. Maximal mass drift $1.3 \cdot 10^{-17}$ (practical) and $6.0 \cdot 10^{-15}$ (tight). (T_E) ratio $\leq 7.9 \cdot 10^{-5}$ (median $6 \cdot 10^{-6}$).

5.6. **Interface evolution: merger and Ostwald ripening.** Protocol: (1) tight solutions for $\tau = 8\tau_0, \ldots, \tau_0$ ($\tau_0 = 2.5 \cdot 10^{-4}$ star, $5 \cdot 10^{-4}$ droplets); (2) N against $2N$; (3) the reported observables; (4) practical and tight solvers in lockstep at the production resolution, same data and output times, constant τ. Observables (Appendix A.7): the number of components of $\{\phi > 0\}$ (periodic, 4-connectivity), a merger time being its first decrease; the interface-displacement proxy $\delta_\Gamma(a, b) = \|a - b\|_{L^1}/(2|\Gamma_b|)$, which equals the mean normal displacement to leading order for equal profiles and nearby interfaces but is not a geometric measurement or bound in general; the equivalent radius $R_j = (\mathcal{A}_j/\pi)^{1/2}$ from the marching-squares zero contour nearest to droplet j's previous centroid (within 0.15), a disappearance time being the first step without such a contour.

TABLE 3. Resolution study (tight solver). Star: δ_Γ against $N = 128$, $\tau = 2.5 \cdot 10^{-4}$ at $t = 0.1, 0.25, 0.5$, and merger times. Droplets: largest difference of the radii at $t = 0.15, 1, 2$ against the run in column 3, and disappearance times t_1, t_2. Production resolutions *.

Star and two circles ($A = 2$, $T = 0.5$): displacement proxy $\delta_\Gamma(t)$ against $N = 128$, $\tau = 2.5 \cdot 10^{-4}$						
N	τ		$\delta_\Gamma(0.1)$	$\delta_\Gamma(0.25)$	$\delta_\Gamma(0.5)$	$t_{m,1}$, $t_{m,2}$
128	0.002		$8.0 \cdot 10^{-4}$	$1.5 \cdot 10^{-3}$	$3.3 \cdot 10^{-4}$	0.03400, 0.1980
128	0.001	*	$1.9 \cdot 10^{-4}$	$3.4 \cdot 10^{-4}$	$7.5 \cdot 10^{-5}$	0.03000, 0.1940
128	0.0005		$3.7 \cdot 10^{-5}$	$6.6 \cdot 10^{-5}$	$1.4 \cdot 10^{-5}$	0.02900, 0.1935
128	0.00025	ref.	–	–	–	0.02875, 0.1930
256	0.00025		$3.1 \cdot 10^{-5}$	$3.4 \cdot 10^{-5}$	$1.6 \cdot 10^{-5}$	0.02850, 0.1930

Unequal droplets ($A = 0$, $T = 2$): radii at $t = 0.15, 1, 2$ against the run in column 3				
N	τ	against	$\max_{j,k} \lvert R_j(t_k) - R_j^{\mathrm{cmp}}(t_k) \rvert$	t_1, t_2
256	0.004	$N = 256$, $\tau = 0.001$	$1.3 \cdot 10^{-3}$	0.2040, 1.5560
256	0.002	$N = 256$, $\tau = 0.001$	$5.3 \cdot 10^{-5}$	0.2000, 1.5540
256	0.001	*	–	0.1990, 1.5520
256	0.0005	$N = 256$, $\tau = 0.001$	$1.3 \cdot 10^{-5}$	0.1985, 1.5520
128	0.0005	$N = 256$, $\tau = 0.0005$	$1.7 \cdot 10^{-4}$	0.1985, 1.5520
512	0.001	$N = 256$, $\tau = 0.001$	$4.2 \cdot 10^{-5}$	0.1990, 1.5520

Merger (Fig. 6): the problem of Fig. 2(c), $[0, 2)^2$, $\varepsilon = 0.04$, $A = 2$, $T = 0.5$; the circles merge at $t \approx 0.029$, the relaxed star joins them through a neck at $t \approx 0.193$. Table 3: δ_Γ decreases at second order in τ; at $N = 128$, $\tau = 10^{-3}$, $\delta_\Gamma \le 3.4 \cdot 10^{-4} \approx h/46$ and the merger times differ from the reference by 1.25τ and τ; $2N$ gives $\delta_\Gamma \le 3.4 \cdot 10^{-5}$. Practical versus tight: identical merger steps (0.030, 0.194); $\delta_\Gamma \le 8.3 \cdot 10^{-8}$ at the output times against $6.1 \cdot 10^{-5}$–$2.8 \cdot 10^{-4}$ for τ versus $\tau/2$ (a factor 700–4600 in δ_Γ); fields differ by $\le 8.8 \cdot 10^{-7}$ in L^2 and $\le 6.1 \cdot 10^{-6}$ pointwise, largest at the neck; 5140 transforms (practical Armijo) against 11180 (tight Newton–PCG), 54.0% fewer. Cost peaks in the first steps and around the second merger, briefly for the tight solver ($t \approx 0.19$–0.24), longer for the practical one ($t \approx 0.17$–0.33).

Ostwald ripening (Fig. 7): $[0, 2)^2$, $\varepsilon = 0.04$, $A = 0$, droplets ($\phi = 1$ inside) of radii $0.16, 0.24, 0.32$ centred at $(1.00, 1.50)$, $(1.45, 0.72)$, $(0.58, 0.62)$, tanh profiles of the periodic distance, closest interfaces $0.32 \approx 8\varepsilon$ apart. The smallest droplet disappears at $t \approx 0.199$, the middle one at $t \approx 1.552$, and $R_3(2) = 0.4195$. At $N = 256$, $\tau = 10^{-3}$ the radii at $t = 0.15, 1, 2$ agree with $\tau/2$ to $1.3 \cdot 10^{-5}$ and with $N = 512$ to $4.2 \cdot 10^{-5}$; disappearance times agree within one step. Practical versus tight: identical disappearance steps; median $\lvert R_j^{\mathrm{p}} - R_j^{\mathrm{t}} \rvert \le 2.4 \cdot 10^{-7}$ against a median time-step effect $\ge 1.5 \cdot 10^{-6}$; maximum $7.4 \cdot 10^{-5}$ (R_2, $t = 1.551$) against a time-step effect of $2.2 \cdot 10^{-3}$ there; $\delta_\Gamma \le 3.4 \cdot 10^{-7}$ at the output times; $\le 4.3 \cdot 10^{-5}$ in L^2 over the run; 14492 transforms (practical Armijo) against 43278 (tight Newton–PCG), 66.5% fewer.

5.7. **Nonzero mean, three seeds.** $[0, 6.4)^2$, $N = 256$, $\varepsilon = 0.04$, $A = 0$, $\tau = 0.002$, $T = 5$, $\phi^0 = \bar\phi + 0.05\, P/\max|P|$ with $\bar\phi = -0.5$ and $P(x) = \sum_{|k_1|,|k_2| \le 24} c_k e^{\mathrm{i}\kappa_k \cdot x}$, $\hat P_0 = c_0 = 0$, $c_{-k} = \overline{c_k}$ (so P is real), $c_k = \frac{1}{2}(z_k + \overline{z_{-k}})$ with $\mathrm{Re}\, z_k, \mathrm{Im}\, z_k$ independent standard normal (NumPy default generator, seeds 1, 2, 3; coefficients stored); $\max|P|$ is taken over the 1024×1024 grid for every N, so the initial function is the same on all grids (the $N = 256$ and $N = 512$ data agree at common points to 10^{-16}). Linearizing (1.1) at $\bar\phi$, $\hat\phi_t = -s(f''(\bar\phi) + \varepsilon^2 s)\hat\phi$ with $f''(\bar\phi) = 3\bar\phi^2 - 1 = -\frac{1}{4}$, so $\lambda_{\max} = \max_s s(\frac{1}{4} - \varepsilon^2 s) = (64\varepsilon^2)^{-1} \approx 9.8$ and $\tau\lambda_{\max} \approx 0.02$.

Refinement (seed 1 only, tight). $N = 512$: L^2 difference $\le 2.3 \cdot 10^{-5}$, $\delta_\Gamma \le 5.1 \cdot 10^{-7}$ at $t = 0.5, 1, 2, 5$, same droplet count at every step. $\tau/2$: L^2 difference $1.3 \cdot 10^{-2}$–$4.8 \cdot 10^{-2}$,

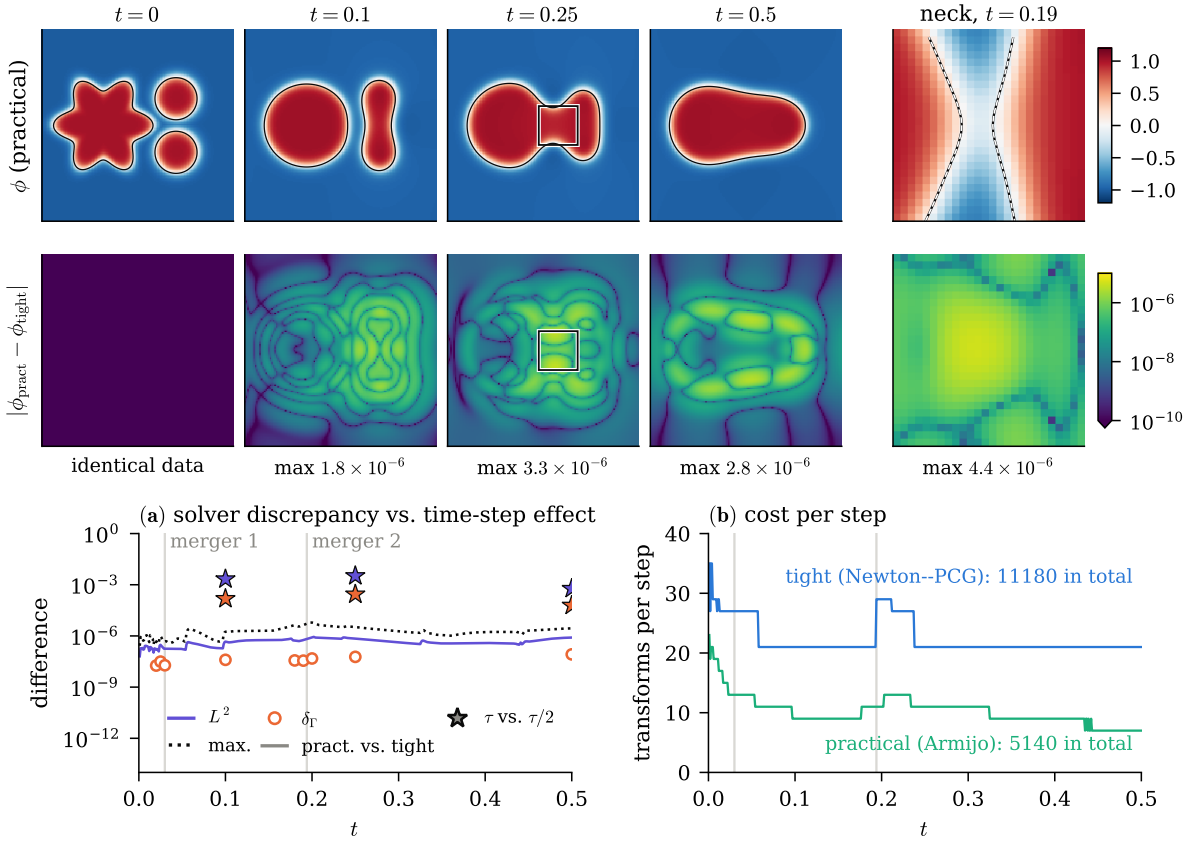


FIGURE 6. Merger ($[0,2)^2$, $N = 128$, $\varepsilon = 0.04$, $A = 2$, $\tau = 10^{-3}$). Top: practical ϕ at $t = 0, 0.1, 0.25, 0.5$, limits $[-1.2, 1.2]$ (values in $[-1.019, 1.044]$), tight zero level set black; column 5: the box marked at $t = 0.25$ (0.41^2, 26^2 points) at $t = 0.19$, practical zero level set white dashed. Bottom: $|\phi_{\text{pract}} - \phi_{\text{tight}}|$ on $[10^{-10}, 10^{-5}]$ (contains every maximum). (a) Practical minus tight: L^2 (line), maximum norm (dotted), δ_Γ (circles); stars: the same L^2 and δ_Γ for tight $\tau = 10^{-3}$ against $5 \cdot 10^{-4}$; vertical lines: merger times. (b) Transforms per step.

TABLE 4. Seed study (practical Armijo, p, against tight Newton–PCG, t): transforms FFT_p / FFT_t and the factor $\text{FFT}_\text{t}/\text{FFT}_\text{p}$; failures; number of steps (of 2500) with different droplet counts (largest difference); largest δ_Γ at $t = 0.5, 1, 2, 5$; largest L^2 difference; largest (T_E) ratio; largest mass drift.

seed	FFT_p / FFT_t	$\text{FFT}_\text{t}/\text{FFT}_\text{p}$	failures	count differs (max)	$\max_k \delta_\Gamma(t_k)$	$\max_n \|\phi_p - \phi_t\|$	(T_E) ratio	mass drift
1	31328 / 74300	2.37	0	4 (1)	$1.6 \cdot 10^{-5}$	$1.0 \cdot 10^{-2}$	$3.1 \cdot 10^{-4}$	$4.9 \cdot 10^{-15}$
2	31190 / 73616	2.36	0	4 (1)	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-2}$	$2.9 \cdot 10^{-4}$	$2.2 \cdot 10^{-15}$
3	30288 / 73752	2.44	0	3 (1)	$1.3 \cdot 10^{-5}$	$3.4 \cdot 10^{-3}$	$3.3 \cdot 10^{-4}$	$6.8 \cdot 10^{-15}$

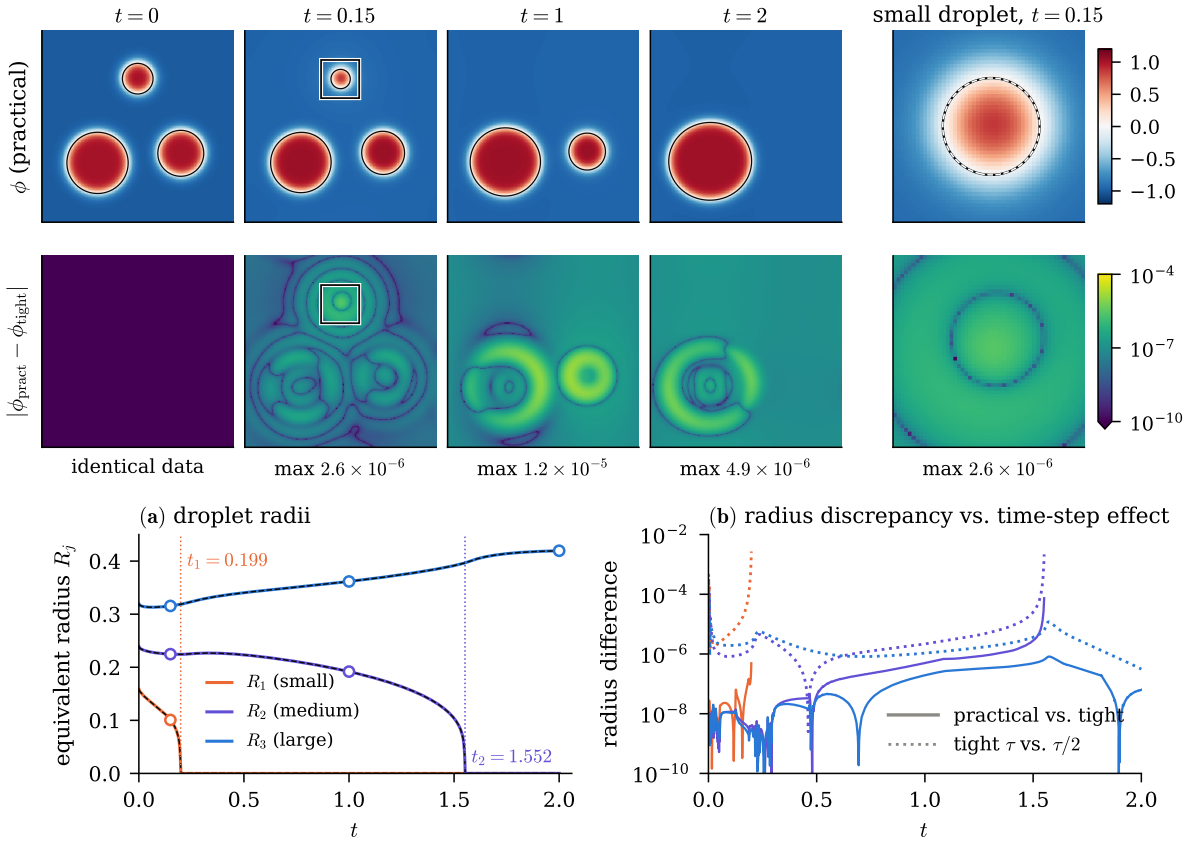


FIGURE 7. Ostwald ripening ($[0,2)^2$, $N = 256$, $\varepsilon = 0.04$, $A = 0$, $\tau = 10^{-3}$). Top: practical ϕ at $t = 0, 0.15, 1, 2$, limits $[-1.2, 1.2]$ (values in $[-1.000, 1.041]$), tight zero level set black; column 5: the small droplet at $t = 0.15$ (box 0.40^2, 51^2 points), practical zero level set white dashed. Bottom: $|\phi_{\text{pract}} - \phi_{\text{tight}}|$ on $[10^{-10}, 10^{-4}]$. (a) $R_j(t)$: practical (coloured), tight (black dashed), tight with $\tau/2$ at the output times (circles); dotted: disappearance times (identical in both runs). (b) $|R_j^{\text{p}} - R_j^{\text{t}}|$ (solid) and $|R_j(\tau) - R_j(\tau/2)|$ of the tight solver (dotted); colours as in (a).

$\delta_\Gamma = 3.0 \cdot 10^{-4}$–$5.2 \cdot 10^{-4}$; counts equal at the output times $(65, 50, 38, 21)$ but different at 148 of 2500 steps (by up to 4). Event timing is therefore not converged to one step at $\tau = 0.002$.

Practical versus tight (all seeds; Fig. 8 and Table 4). No failures (practical ≤ 9 iterations, tight ≤ 4 Newton iterations, cap 60); $\widetilde{E}$ decreases at every step of all six runs; counts rise from 0 to 67–76 at $t \approx 0.35$ and coarsen to 21–23; counts differ at 3–4 of 2500 steps, by one, against 148 for $\tau/2$; $\delta_\Gamma \leq 1.6 \cdot 10^{-5}$ at the output times (20–63 times less than the $\tau/2$ effect, seed 1); the largest L^2 differences ($1.0 \cdot 10^{-2}$, $1.0 \cdot 10^{-2}$, $3.4 \cdot 10^{-3}$ at $t = 2.68, 2.75, 2.86$) occur at or one step after a droplet removal that happens one step earlier in one run, the median being below 10^{-3}; 57.6–58.9% fewer transforms (practical Armijo against tight Newton–PCG).

5.8. **Synthesis and concluding remarks.** Table 5 collects the lockstep comparisons: the practical rule (Armijo) used 54–67% fewer transforms than tight Newton–PCG, reproduced merger and disappearance steps, matched droplet radii to $7.4 \cdot 10^{-5}$ and droplet counts at the output times, and kept $\widetilde{E}$ decreasing, as Theorem 3.12 guarantees. The unrelaxed iteration

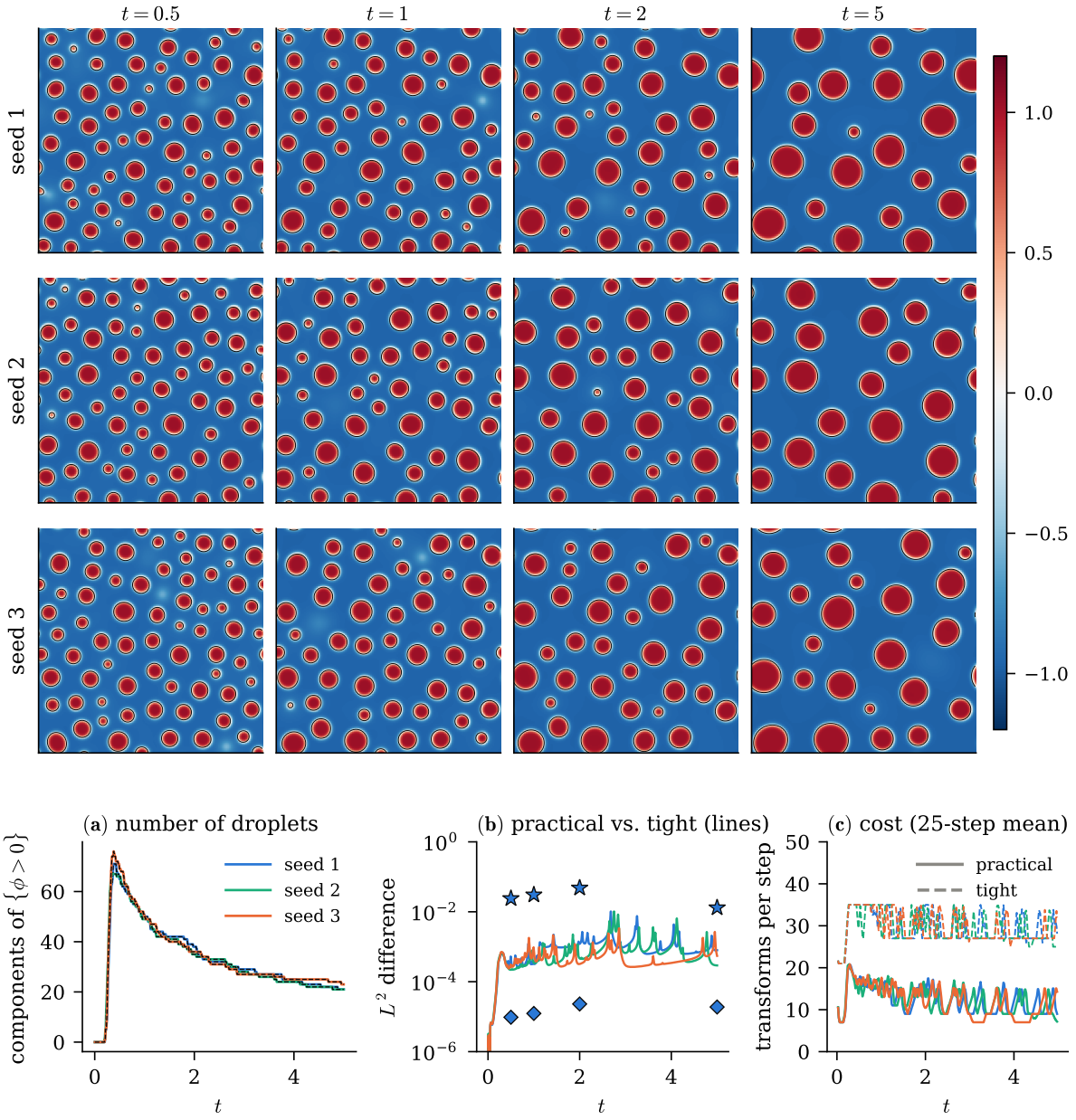


FIGURE 8. Nonzero mean ($[0, 6.4)^2$, $N = 256$, $\varepsilon = 0.04$, $A = 0$, $\tau = 0.002$, $\bar{\phi} = -0.5$). Top: practical ϕ for seeds 1–3 at $t = 0.5, 1, 2, 5$, limits $[-1.2, 1.2]$ (values in $[-0.980, 1.039]$), tight zero level set black. (a) Components of $\{\phi > 0\}$: practical coloured, tight black dashed. (b) L^2 difference practical–tight (lines, colours as in (a)); seed 1, tight: τ against $\tau/2$ (stars), N against $2N$ (diamonds). (c) Transforms per step, 25-step moving average: practical solid, tight dashed.

TABLE 5. Practical rule with Armijo inner method (p) versus tightly converged Newton–PCG (t) on all lockstep runs: steps, total transforms, saving, topological events (merger or disappearance steps; for the seeds, steps with different droplet counts), largest L^2 difference, largest (T_E) ratio and mass drift. $\widetilde{E}$ decreased at every step of every run.

run	steps	FFT_p	FFT_t	fewer	events	$\max_n \lVert\phi^n_\mathrm{p} - \phi^n_\mathrm{t}\rVert$	(T_E) ratio	mass drift
labyrinth, 512^2, $\tau = 0.004$	2500	43892	112406	61.0%	–	$4.7 \cdot 10^{-3}$	$7.9 \cdot 10^{-5}$	$6.0 \cdot 10^{-15}$
merger, 128^2, $\tau = 10^{-3}$	500	5140	11180	54.0%	same	$8.8 \cdot 10^{-7}$	$6.3 \cdot 10^{-6}$	$1.7 \cdot 10^{-15}$
ripening, 256^2, $\tau = 10^{-3}$	2000	14492	43278	66.5%	same	$4.3 \cdot 10^{-5}$	$6.3 \cdot 10^{-5}$	$3.7 \cdot 10^{-15}$
$\bar\phi = -0.5$, seed 1, 256^2, $\tau = 0.002$	2500	31328	74300	57.8%	4/2500	$1.0 \cdot 10^{-2}$	$3.1 \cdot 10^{-4}$	$4.9 \cdot 10^{-15}$
$\bar\phi = -0.5$, seed 2, 256^2, $\tau = 0.002$	2500	31190	73616	57.6%	4/2500	$1.0 \cdot 10^{-2}$	$2.9 \cdot 10^{-4}$	$2.2 \cdot 10^{-15}$
$\bar\phi = -0.5$, seed 3, 256^2, $\tau = 0.002$	2500	30288	73752	58.9%	3/2500	$3.4 \cdot 10^{-3}$	$3.3 \cdot 10^{-4}$	$6.8 \cdot 10^{-15}$

diverged in regimes not covered by Corollary 3.3, and fixed relaxed sweeps lost order, consistent with (3.21) and Proposition 3.15; globalized methods stopped by Algorithm 1 avoided both.

The arguments of Sections 2 and 3 use only convexity of the implicit part and the constant-coefficient preconditioner, and should extend to other secant and convex-splitting schemes, BDF2 variants and finite-difference discretizations. Error bounds polynomial in ε^{-1} and logarithmic potentials, where maximum-norm control must be replaced by separation from the singular values, need further ideas.

Code and data. The archive `ch_inner_iterations_submission.zip` contains the solver (`code/chsolver.py`), the simulation scripts (`experiments.py`, `fig_experiments.py`, `fig_experiments2.py`), the raw data (`data/`), the diagnostics (`diagnostics.py`, which writes every number quoted in Section 5 to `data/summary.json`) and the plotting script. From the stored data, all figures and tables are regenerated by `python3 diagnostics.py` followed by `python3 make_figures.py 1 2 3 4 5 6 7 S2 tables`; the commands that rerun each simulation are listed in `code/README.md`.

APPENDIX A. AUXILIARY PROOFS AND IMPLEMENTATION

A.1. Discrete Sobolev inequality.

Proof of Lemma 2.1(c). For $f \in \mathring{\mathcal{V}}$ and every grid point,

$$|f_{ij}| \le \sum_{k\neq 0} |\hat f_k| \le \Big(\sum_k s_k |\hat f_k|^2\Big)^{1/2} \Big(\sum_{k\neq 0} s_k^{-1}\Big)^{1/2} = \frac{\|\nabla_N f\|}{L} \cdot \frac{L}{2\pi} \Big(\sum_{k\neq 0} |k|^{-2}\Big)^{1/2}.$$

The shell $\max(|k_1|,|k_2|) = j$ contains at most $8j$ indices, each with $|k| \ge j$, so

$$\sum_{k\in K_N\setminus\{0\}} |k|^{-2} \le \sum_{j=1}^{N/2} \frac{8j}{j^2} = 8\sum_{j=1}^{N/2} \frac{1}{j} \le 8\Big(1 + \ln\frac{N}{2}\Big) \le 8(1+\ln N),$$

hence $\|f\|_\infty \le \frac{\sqrt{8}}{2\pi}(1+\ln N)^{1/2}\|\nabla_N f\| = C_N\|\nabla_N f\|$. In three dimensions the shell has at most $24j^2 + 2$ indices and the sum is $O(N)$. □

A.2. Discrete Gronwall inequality.

Lemma A.1. *If $y_k \ge 0$ and $y_k \le F + \Lambda\tau \sum_{j=0}^{k-1} y_j$ for $0 \le k \le K$, then $y_k \le F(1+\Lambda\tau)^k \le F e^{\Lambda t_k}$.*

Proof. $Z_k = F + \Lambda\tau\sum_{j<k} y_j$ satisfies $Z_0 = F$ and $Z_{k+1} = Z_k + \Lambda\tau y_k \le (1+\Lambda\tau)Z_k$; and $y_k \le Z_k$. □

A.3. **Projection and interpolation estimates.**

Lemma A.2. *For L-periodic g and $m \ge 1$, with C depending on L and m only,*

$$\|I_N g\|_{H^1} \le C\|g\|_{H^2}, \qquad \|g - \mathsf{P}_N g\|_{H^1} + \|g - I_N g\|_{H^1} \le C h^m \|g\|_{H^{m+1}}.$$

Proof. Scale $L = 2\pi$ and write $\langle q\rangle = (1+|q|^2)^{1/2}$, $q \in \mathbb{Z}^2$. The interpolant has $\widehat{I_N g}_k = \sum_{\ell\in\mathbb{Z}^2} \hat g_{k+N\ell}$, $k \in K_N$. For $k \in K_N$ and $\ell \ne 0$, $|k+N\ell| \ge N|\ell|_\infty - \frac{N}{2} \ge \frac{N}{2}|\ell|_\infty$, so by weighted Cauchy–Schwarz

$$\langle k\rangle^2 \Big|\sum_\ell \hat g_{k+N\ell}\Big|^2 \le \Big\{\langle k\rangle^2 \sum_\ell \langle k+N\ell\rangle^{-4}\Big\}\sum_\ell \langle k+N\ell\rangle^4 |\hat g_{k+N\ell}|^2,$$
$$\Big\{\langle k\rangle^2 \sum_\ell \langle k+N\ell\rangle^{-4}\Big\} \le 1 + \big(1+\tfrac{N^2}{2}\big)\tfrac{16}{N^4}\sum_{\ell\ne 0}|\ell|_\infty^{-4} \le C.$$

Summing over the disjoint classes $k + N\mathbb{Z}^2$ gives the first bound. Next, $g - \mathsf{P}_N g$ contains only $|q|_\infty \ge \frac{N}{2}$, so $\|g-\mathsf{P}_N g\|_{H^1}^2 \le (\frac{N}{2})^{-2m}\sum\langle q\rangle^{2m+2}|\hat g_q|^2$. Since $I_N\mathsf{P}_N g = \mathsf{P}_N g$, $g - I_N g = (g-\mathsf{P}_N g) - I_N(g-\mathsf{P}_N g)$, and with exponent $m+1$,

$$\|I_N(g-\mathsf{P}_N g)\|_{H^1}^2 \le \sum_{k\in K_N}\Big\{\langle k\rangle^2 \sum_{q\equiv k,\,|q|_\infty\ge N/2}\langle q\rangle^{-2(m+1)}\Big\}\sum_{q\equiv k}\langle q\rangle^{2(m+1)}|\hat g_q|^2 \le CN^{-2m}\|g\|_{H^{m+1}}^2,$$

because $\{\cdot\} \le (1+\frac{N^2}{2})(\frac{N}{2})^{-2(m+1)}\sum_\ell \max(1,|\ell|_\infty)^{-2(m+1)} \le CN^{-2m}$ for $m \ge 1$. □

A.4. **Proof of Lemma 4.2.** Let $\bar t = \bar t_n = t_{n+1/2}$, $u_j = u(t_j)$, and for $g \in C^3$ use $g(\bar t+\vartheta) = g + \vartheta g' + \frac{\vartheta^2}{2}g'' + \mathcal{T}_3(\vartheta)$, $\mathcal{T}_3(\vartheta) = \frac12\int_{\bar t}^{\bar t+\vartheta}(\bar t+\vartheta-r)^2 g'''(r)\,dr$, $|\mathcal{T}_3(\vartheta)| \le \frac{|\vartheta|^3}{6}\sup|g'''|$. With $t_{n+1} = \bar t + \frac{\tau}{2}$, $t_n = \bar t - \frac{\tau}{2}$, $t_{n-1} = \bar t - \frac{3\tau}{2}$, the coefficients of $\vartheta^0, \vartheta^1, \vartheta^2$ give

$$\begin{aligned}
\tfrac{1}{\tau}(u_{n+1}-u_n) - u_t(\bar t) &= \tfrac{1}{\tau}\big[\mathcal{T}_3(\tfrac{\tau}{2}) - \mathcal{T}_3(-\tfrac{\tau}{2})\big], && |\cdot| \le \tfrac{\tau^2}{24}\sup|u_{ttt}|,\\
c_1 := \tfrac34 u_{n+1} + \tfrac14 u_{n-1} - u(\bar t) &= \tfrac38\tau^2 u_{tt}(\bar t) + \varrho_1, && |\varrho_1| \le \tfrac{5}{32}\tau^3\sup|u_{ttt}|,\\
c_2 := \tfrac32 u_n - \tfrac12 u_{n-1} - u(\bar t) &= -\tfrac38\tau^2 u_{tt}(\bar t) + \varrho_2, && |\varrho_2| \le \tfrac{5}{16}\tau^3\sup|u_{ttt}|,\\
c_3 := u_{n+1} - 2u_n + u_{n-1} &= \tau^2 u_{tt}(\bar t) + \varrho_3, && |\varrho_3| \le \tfrac58\tau^3\sup|u_{ttt}|,
\end{aligned}$$

since, e.g. for c_1: $\frac34\cdot\frac{\tau}{2} + \frac14\cdot(-\frac{3\tau}{2}) = 0$, $\frac12(\frac34\cdot\frac{\tau^2}{4} + \frac14\cdot\frac{9\tau^2}{4}) = \frac38\tau^2$ and $\frac16(\frac34\cdot\frac18 + \frac14\cdot\frac{27}{8})\tau^3 = \frac{5}{32}\tau^3$; similarly for c_2 ($-\frac34\tau + \frac34\tau = 0$, $\frac12(\frac38 - \frac98)\tau^2$) and c_3 ($\frac{\tau}{2} + \tau - \frac{3\tau}{2} = 0$, $\frac12(\frac14 - \frac12 + \frac94)\tau^2$). For the secant, with $\bar u = \frac12(u_{n+1}+u_n) = u + \frac{\tau^2}{8}u_{tt} + O(\tau^3)$ and $d = \frac12(u_{n+1}-u_n) = \frac{\tau}{2}u_t + O(\tau^3)$,

$$\chi(u_{n+1},u_n) = \tfrac14(2\bar u^2 + 2d^2)(2\bar u) = \bar u^3 + \bar u d^2 = u^3 + \tau^2\big(\tfrac38 u^2 u_{tt} + \tfrac14 u u_t^2\big) + O(\tau^3)$$

in H^{m+2}, which is an algebra. Let $\tilde\mu = \chi(u_{n+1},u_n) - \frac32 u_n + \frac12 u_{n-1} - \varepsilon^2\Delta(\frac34 u_{n+1} + \frac14 u_{n-1}) + Ac_3$. Then

$$\tilde\mu - \mu(u(\bar t)) = (\chi - u^3) - c_2 - \varepsilon^2\Delta c_1 + Ac_3 = \tau^2 W(\bar t) + O(\tau^3) \quad \text{in } H^{m+2}. \tag{A.1}$$

Since $u_t = \Delta\mu(u)$, $\mathsf{P}_N\Delta = \Delta\mathsf{P}_N$, and $\Delta_N = \Delta$ on grid values of P_N-images (no Nyquist modes),

$$\zeta^n = \zeta_1^n - \Delta\mathsf{P}_N\big[\tilde\mu - \mu(u(\bar t))\big] + \sigma^n, \qquad \sigma^n = \Delta\mathsf{P}_N g - \Delta_N\chi(U^{n+1},U^n), \quad g = \chi(u_{n+1},u_n),$$

the linear parts of $\tilde{\mu}$ cancelling exactly. With $\|\Delta w\|_{-1} = \|\nabla w\|$, (A.1) gives the $\tau^2 W$ term and $O(\tau^3)$. For σ^n identify grid functions with trigonometric interpolants I_N, so that $\|\nabla_N w\| = \|\nabla I_N w\|$, and write

$$\sigma^n = \Delta(\mathsf{P}_N g - I_N g) - \Delta I_N\big[\chi(\mathsf{P}_N u_{n+1}, \mathsf{P}_N u_n) - g\big].$$

By Lemma A.2 (see also [3, §5.1], [17]),

$$\|\sigma^n\|_{-1} \le Ch^m\|g\|_{H^{m+1}} + C\|\chi(\mathsf{P}_N u_{n+1}, \mathsf{P}_N u_n) - g\|_{H^2} \le Ch^m\big(\|g\|_{H^{m+1}} + \|u\|_{L^\infty(H^{m+2})}\big).$$

For $n = 0$, u_{-1} is replaced by u_0, and $c_1 = \frac{3}{4}(u_1 - u(\bar{t})) + \frac{1}{4}(u_0 - u(\bar{t}))$, $c_2 = u_0 - u(\bar{t})$, $c_3 = u_1 - u_0$ are $O(\tau)$. Finally $\operatorname{mean}(U^{n+1} - U^n) = 0$ by conservation of mass and $\operatorname{mean}(\Delta_N \cdot) = 0$, so $\zeta^n \in \mathring{\mathcal{V}}$. □

A.5. **Proof of Proposition 3.15.** At $\phi = 0$, $\partial_a\chi = \partial_b\chi = 0$, so the linearised step on the mode s reads, with $\mu = -(\frac{3}{2}a_n - \frac{1}{2}a_{n-1}) + \varepsilon^2 s(\frac{3}{4}a_* + \frac{1}{4}a_{n-1}) + A(a_* - 2a_n + a_{n-1})$ and $a_* - a_n = -\tau s\mu$,

$$\begin{aligned} a_* &= P(\tau)a_n + Q(\tau)a_{n-1}, \qquad d_s = As + \tfrac{3}{4}\varepsilon^2 s^2, \\ P &= \frac{1 + \tau(2A + \frac{3}{2})s}{1 + \tau d_s}, \qquad Q = -\frac{\tau\{(A + \frac{1}{2})s + \frac{1}{4}\varepsilon^2 s^2\}}{1 + \tau d_s}. \end{aligned} \tag{A.2}$$

The linearised map $\mathcal{G}_n$ does not depend on its argument (χ contributes nothing), so $\mathcal{G}_n(v) = a_*$ and K relaxed sweeps from $v^0 = 2a_n - a_{n-1}$ give $v^K - a_* = r(v^0 - a_*)$, i.e.

$$a_{n+1} = (1 - r)a_* + r(2a_n - a_{n-1}).$$

With $a_n = g^n$ the characteristic equation is

$$F(g, \tau) = g^2 - \{(1 - r)P(\tau) + 2r\}g - (1 - r)Q(\tau) + r = 0, \qquad F(g, 0) = (g - 1)(g - r). \tag{A.3}$$

Since $0 \le r < 1$ the roots at $\tau = 0$ are simple, and the implicit function theorem gives analytic branches through 1 and r. With $p_s = (A + \frac{3}{2})s - \frac{3}{4}\varepsilon^2 s^2$ and $q_s = -(A + \frac{1}{2})s - \frac{1}{4}\varepsilon^2 s^2$,

$$P = 1 + p_s\tau - d_s p_s\tau^2 + O(\tau^3), \qquad Q = q_s\tau - d_s q_s\tau^2 + O(\tau^3), \qquad p_s + q_s = \lambda, \qquad p_s - d_s = \tfrac{3}{2}\lambda.$$

Inserting $g = 1 + \alpha\tau + \gamma\tau^2 + O(\tau^3)$ into (A.3), the coefficient of τ^0 vanishes, that of τ is $(1 - r)\alpha - (1 - r)(p_s + q_s)$, and that of τ^2 is

$$\begin{aligned} &(1 - r)\gamma + \alpha^2 - (1 - r)\alpha p_s + (1 - r)d_s(p_s + q_s) \\ &\quad = (1 - r)\gamma + \lambda^2 - (1 - r)\lambda(p_s - d_s) = (1 - r)\gamma + \lambda^2 - \tfrac{3}{2}(1 - r)\lambda^2. \end{aligned}$$

Hence $\alpha = \lambda$, $\gamma = \frac{3}{2}\lambda^2 - \frac{\lambda^2}{1-r} = (\frac{1}{2} - \frac{r}{1-r})\lambda^2$, and $\log(1 + \alpha\tau + \gamma\tau^2) = \alpha\tau + (\gamma - \frac{1}{2}\alpha^2)\tau^2 + O(\tau^3) = \lambda\tau - \frac{r}{1-r}\lambda^2\tau^2 + O(\tau^3)$. Raising to the power T/τ gives the last claim. □

A.6. **Fourier-space formulas.** With $\delta = v - \phi^n$, $\delta_2 = v - 2\phi^n + \phi^{n-1}$, $\hat{\chi} = \widehat{\chi(v, \phi^n)}$ (one forward transform), and sums over $k \ne 0$:

$$\hat{p}_k = \hat{v}_k - \lambda_k^{-1}(\hat{f}_{n,k} - s_k\hat{\chi}_k), \qquad \widehat{\mathcal{R}_n(v)}_k = \frac{\lambda_k}{s_k}\hat{p}_k, \qquad \hat{f}_{n,k} = \tfrac{1}{\tau}\hat{\phi}^n_k + s_k\hat{\ell}_{n,k} - \tfrac{1}{4}\varepsilon^2 s_k^2\hat{\phi}^{n-1}_k,$$

$$\|\mathcal{R}_n\|^2_{\mathcal{B}} = L^2\sum\frac{\lambda_k}{s_k}|\hat{p}_k|^2, \qquad \|\mathcal{R}_n\|^2 = L^2\sum\frac{\lambda_k^2}{s_k^2}|\hat{p}_k|^2, \qquad (\mathcal{R}_n, \delta) = L^2\sum\frac{\lambda_k}{s_k}\operatorname{Re}\big(\hat{p}_k\overline{\hat{\delta}_k}\big),$$

$$\mathcal{D}_n(v) = L^2\sum\Big[\frac{|\hat{\delta}_k|^2}{\tau s_k} + \Big(\tfrac{1}{4} + \tfrac{A}{2} + \tfrac{\varepsilon^2}{8}s_k\Big)|\hat{\delta}_{2,k}|^2\Big], \qquad \|d\|^2_{\mathcal{B}^{-1}} = L^2\sum\frac{\lambda_k}{s_k}|\hat{d}_k|^2.$$

The relaxed and Armijo updates are $\hat{v} \leftarrow \hat{v} - \omega\hat{p}$, followed by one inverse transform; the Armijo trial values use (3.9) with $d = -\omega p$, i.e. $(\mathcal{R}_n, d) = -\omega\|\mathcal{R}_n\|^2_{\mathcal{B}}$, $\frac{1}{2}\|d\|^2_{\mathcal{B}^{-1}} = \frac{\omega^2}{2}\|\mathcal{R}_n\|^2_{\mathcal{B}}$ and a

pointwise quartic sum, without transforms. A Newton–CG product $Hw = w + \mathcal{B}\mathsf{P}_0(\chi_a(v,\phi^n)w)$ costs one transform pair: $\widehat{Hw}_k = \hat{w}_k + \frac{s_k}{\lambda_k}\widehat{(\chi_a w)}_k$, $k \neq 0$. For real FFTs the sums over K_N are taken with weight 2 on the non-self-conjugate half-plane columns.

A.7. **Geometric observables.** *Components.* $\{\phi > 0\}$ is labelled with 4-connectivity on the $N \times N$ array; labels touching across $i = 0 \leftrightarrow N-1$ or $j = 0 \leftrightarrow N-1$ are merged (union–find).

Displacement proxy. Let $\psi(r) = \tanh(r/(\sqrt{2}\varepsilon))$, and let $b = \psi(r)$, $a = \psi(r - \nu(\sigma))$ in normal coordinates (σ, r) near Γ_b, $|\nu|$ small. Then $a - b = -\nu(\sigma)\psi'(r) + O(\nu^2\psi'')$ and, with $dx = (1 + \kappa r)\, dr\, d\sigma$,

$$\|a-b\|_{L^1} = \int_{\Gamma_b} |\nu| \int_{\mathbb{R}} |\psi'(r)|(1+\kappa r)\, dr\, d\sigma + O\big(\varepsilon^{-1}\|\nu\|_\infty^2\big) = 2\int_{\Gamma_b} |\nu|\, d\sigma + O\big(\varepsilon^{-1}\|\nu\|_\infty^2\big),$$

since $\int_{\mathbb{R}} \psi' = \psi(\infty) - \psi(-\infty) = 2$, $\int_{\mathbb{R}} r\psi'(r)\, dr = 0$ (ψ' even) and $\int_{\mathbb{R}} |\psi''| = O(\varepsilon^{-1})$. Hence $\delta_\Gamma(a,b) = |\Gamma_b|^{-1}\int_{\Gamma_b} |\nu|\, d\sigma$ to leading order. For a rigid shift z of circles, $\nu = z \cdot n$ and $|\Gamma|^{-1}\int |z \cdot n| = |z|\frac{1}{2\pi}\int_0^{2\pi} |\cos\vartheta|\, d\vartheta = \frac{2}{\pi}|z|$; for three droplets and $|z| = 0.01$ the code returns 0.00636 against $\frac{2}{\pi}\cdot 0.01 = 0.00637$. Bulk differences ($|a|, |b| \approx 1$ away from Γ) and profile changes add to $\|a-b\|_{L^1}$, which is why δ_Γ is used as a defined proxy only. $|\Gamma_b|$ is the length of the marching-squares zero contour.

Radii. For each closed marching-squares zero contour with vertices (x_i, y_i),

$$A_s = \tfrac{1}{2}\sum_i c_i, \quad \mathcal{A} = |A_s|, \quad \text{centroid} = \tfrac{1}{6A_s}\sum_i (x_i + x_{i+1},\, y_i + y_{i+1})\, c_i, \quad c_i = x_i y_{i+1} - x_{i+1} y_i.$$

Droplet j is matched to the closed contour whose centroid is nearest to its previous centroid, if closer than 0.15; then $R_j = (\mathcal{A}/\pi)^{1/2}$ and the centroid is updated; otherwise the droplet is recorded as disappeared at that step.

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[Affiliations]
Email address: [e-mail]